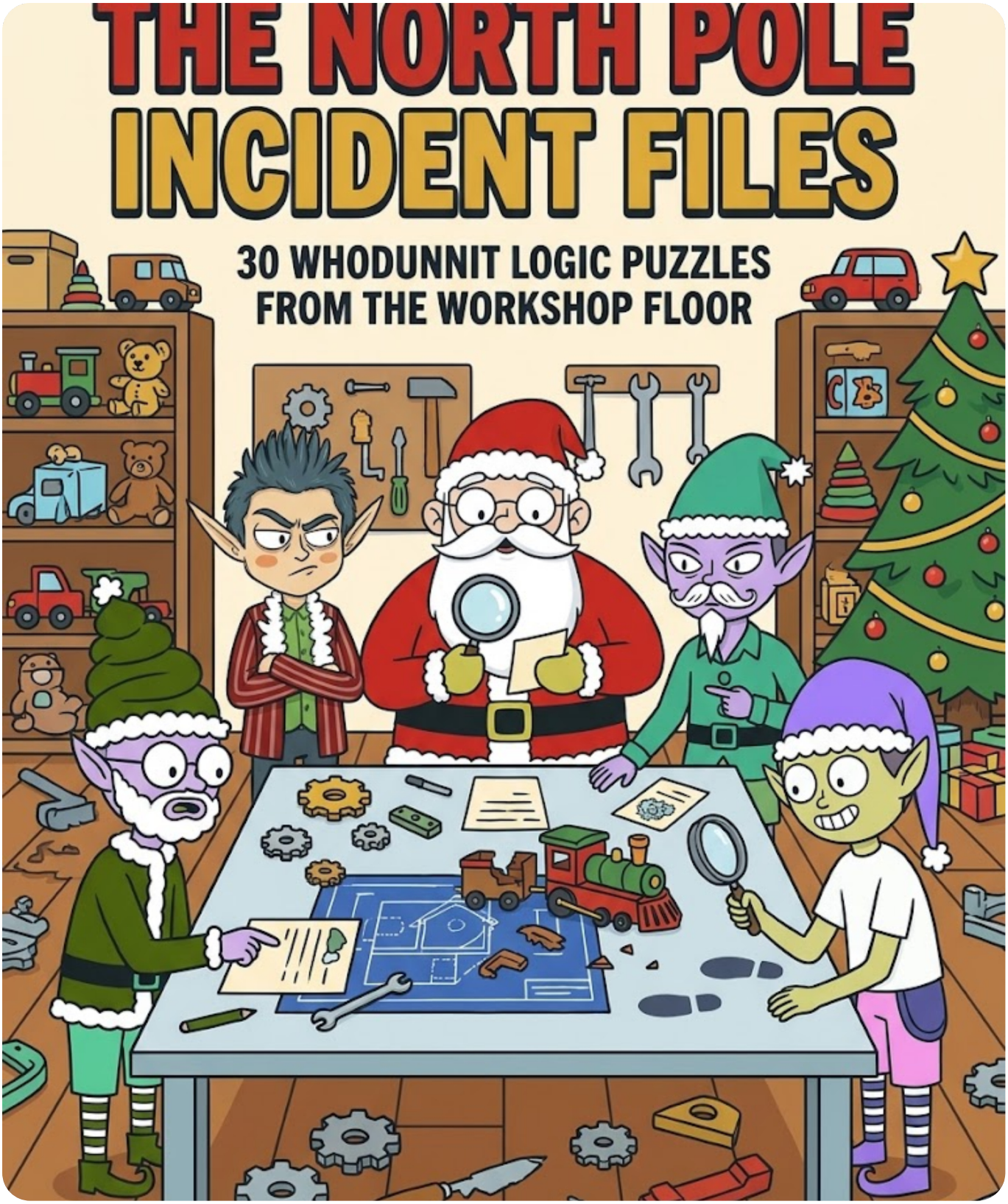


The North Pole Incident Files

30 whodunnit logic puzzles from the workshop floor

A fan-made puzzle booklet for the SANS Holiday Hack Challenge community

THE STORY SO FAR

Prologue: The Suggestion Box

Day one. You are the newest member of the North Pole security team, which until this morning consisted of a suggestion box. Santa welcomes you with a mug of cocoa and a problem. Small things keep going wrong on the workshop floor: a stray USB stick here, an unlocked screen there. 'Probably nothing,' he says, in the voice of a man who has said 'probably nothing' before.

Then he shows you the note. It was found frozen to the door of the sleigh vault, written in ice-blue ink: 'I have changed the password of your vault. It has thirty characters. You will find one under the feet of every fool who helps me. Catch all thirty, or the sleigh stays on the ground. Merry Christmas.' The note is unsigned. It smells of peppermint and smugness. The security team, meaning you, has no leads.

LEVEL 1

Recruit

CASES	MAP	SUSPECTS	NEW
01 - 06	5 × 5	4	One person at a time

How to solve this level

Work out where everybody stood. For each person, note the squares their clues allow. In this level somebody is always down to exactly one square: place that person, cross out their row and column for everybody else, and look again. When all the people are placed, the target is on the one square whose row and column are still free. Whoever shares the target's room is the culprit. Guide: chapter 6.

The vault password so far

Write down the character from every solved case. All thirty have to be right.

LEVEL 1						LEVEL 2						LEVEL 3						LEVEL 4						LEVEL 5					
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30

CASE 01

LEVEL 1 · RECRUIT

The Stray USB Stick

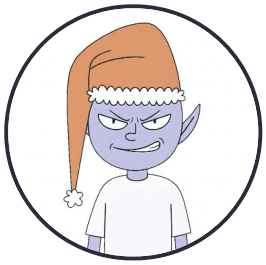
It was lying on the floor in plain view: a USB stick with a snowflake sticker and a handwritten label reading 'BONUS LIST - CONFIDENTIAL'. Every security briefing since the invention of the sleigh has covered this exact situation. The stick lasted fifty-three minutes before somebody picked it up and plugged it into a computer, 'just to see whose it is'. It is now blinking happily and typing commands all by itself. Santa would like a word with whoever could not resist.

TARGET
the USB stick

Suspects and clues

Jewel Loggins (JL)

1. Jewel Loggins was in the north half of the floor.



Poinsettia Mcmittens (PO)

2. Poinsettia Mcmittens was right next to a filing cabinet.
3. Poinsettia Mcmittens was in the west half of the floor.

Jack Frost (JF)

4. Jack Frost was by the east outer wall.
5. Jack Frost was in the south half of the floor.

Santa (SA)

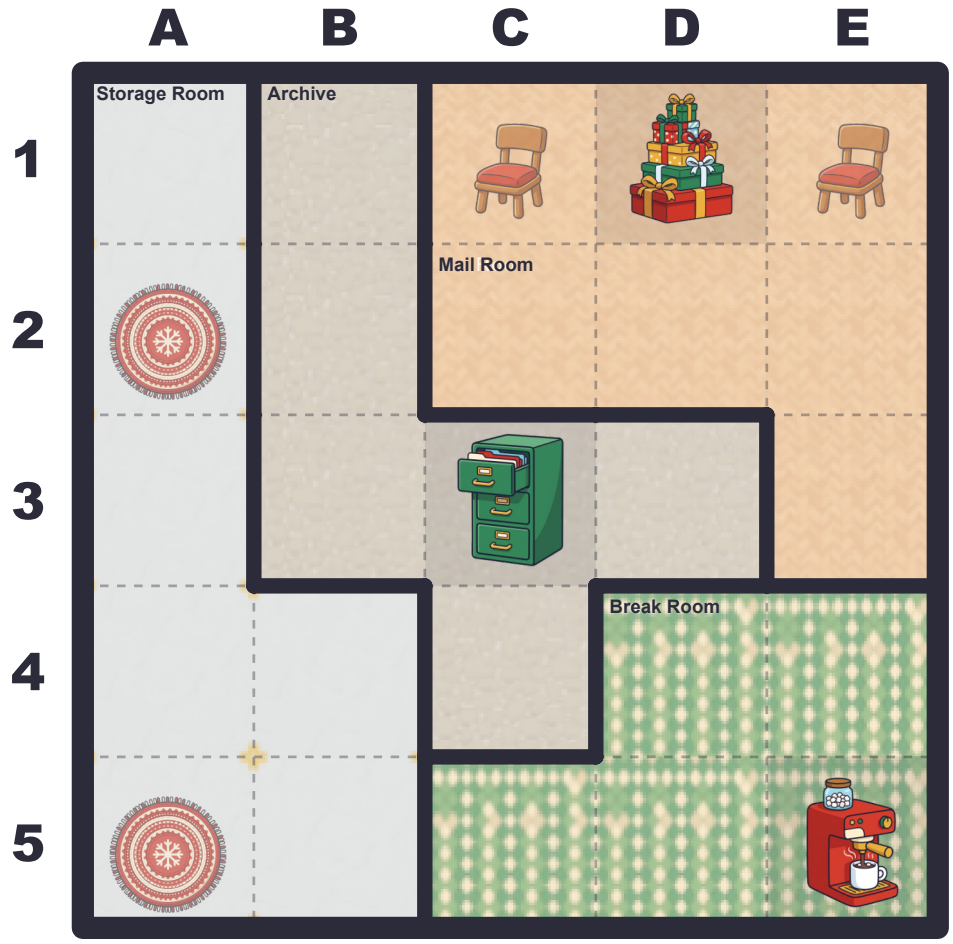
6. Santa was in the north half of the floor.
7. Santa was on a rug.

CASE 01

The Stray USB Stick

Map of the floor · 5 × 5 · north is up

N



ROOMS

Archive · Break Room · Storage Room · Mail Room

BLOCKED: NOBODY STANDS HERE

gift pile · filing cabinet · cocoa machine

CAN BE STOOD ON

chair · rug

Notes

WHO	POSSIBLE SQUARES
JL Jewel Loggins	
PO Poinsettia Mcmittens	
JF Jack Frost	
SA Santa	
T target	

Code grid

	A	B	C	D	E
1	F	f	f	g	S
2	n	f	y	i	2
3	G	L	3	r	M
4	r	B	j	V	f
5	A	f	X	t	Y

The character on the culprit's square is character **1** of the password.

Culprit _____ **No. 1** ☐

CASE 02

LEVEL 1 · RECRUIT

The Unlocked Screen

Santa stepped away from his laptop for 'two minutes' to settle an argument about the correct temperature of cocoa. Two minutes at the North Pole last about as long as two minutes anywhere else when cocoa is involved, which is to say twenty. He did not lock the screen. When he came back, the wish list was open and scrolled all the way down to the letter W, and Alabaster Snowball swears the mug beside the keyboard was still warm. Somebody sat down. Somebody had a good long look.

TARGET
Santa's laptop

Suspects and clues

Santa — SA

1. Santa was by the north outer wall.
2. Santa was by the east outer wall.

Morcel Nougat — MN

3. Morcel Nougat was in the south half of the floor.
4. Morcel Nougat was in a room with a Christmas tree.

Alabaster Snowball — AS

5. Alabaster Snowball was in the south half of the floor.
6. Alabaster Snowball was in the west half of the floor.

Jack Frost — JF

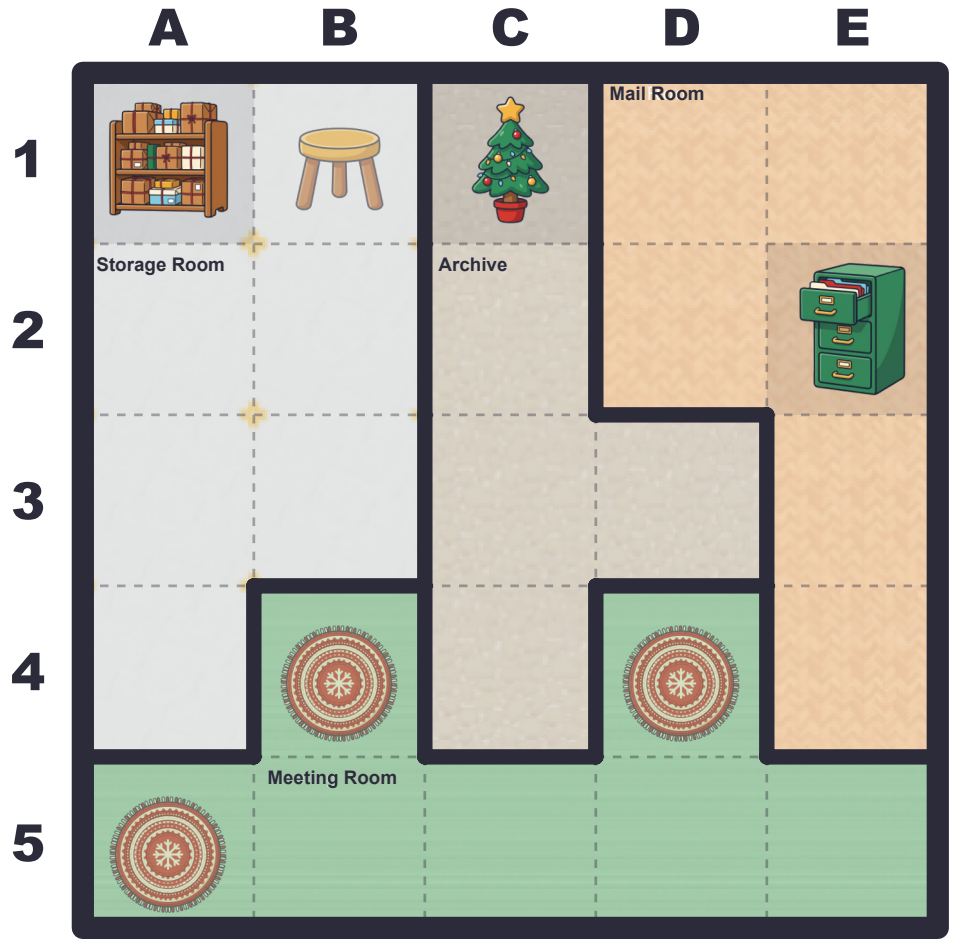
7. Jack Frost was right next to a parcel shelf.

CASE 02

The Unlocked Screen

Map of the floor · 5 × 5 · north is up

N



ROOMS

Archive · Mail Room · Storage Room · Meeting Room

BLOCKED: NOBODY STANDS HERE

parcel shelf · Christmas tree · filing cabinet

CAN BE STOOD ON

stool · rug

Notes

WHO	POSSIBLE SQUARES
SA Santa	
MN Morcel Nougat	
AS Alabaster Snowball	
JF Jack Frost	
T target	

Code grid

	A	B	C	D	E
1	T	r	T	2	4
2	e	6	F	Y	A
3	e	t	N	W	V
4	j	5	e	Y	Y
5	2	V	M	2	n

The character on the culprit's square is character **2** of the password.

Culprit _____ **No. 2** ☐

CASE 03

LEVEL 1 · RECRUIT

The Sticky Note

The password for the gift database is twenty-four characters long, changes every month and must never be written down. So naturally it was found written down: on a bright yellow sticky note, in block capitals, underlined twice, with a helpful little arrow labelled 'gift database'. Somebody on this floor has decided that remembering things is a job for reindeer. The note has since been photographed by at least one person who should not own a camera.

TARGET
the sticky note

Suspects and clues

Jack Frost (JF)

1. Jack Frost was right next to a gift pile.

Alabaster Snowball (AS)

2. Alabaster Snowball was in the south half of the floor.
3. Alabaster Snowball was in the Print Room.

Santa (SA)

4. Santa was in the Print Room.

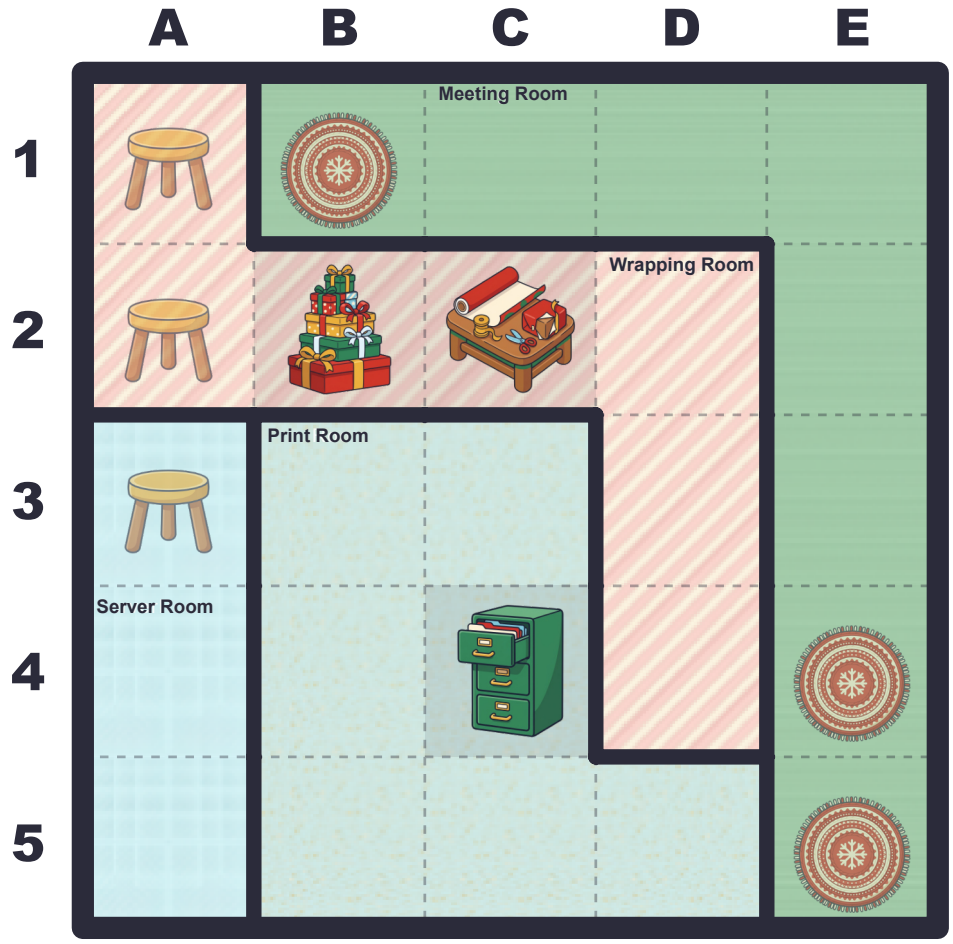
Jewel Loggins (JL)

5. Jewel Loggins was by the south outer wall.
6. Jewel Loggins was in the Meeting Room.

The Sticky Note

Map of the floor · 5 × 5 · north is up

N



ROOMS

Wrapping Room · Server Room · Print Room · Meeting Room

BLOCKED: NOBODY STANDS HERE

gift pile · wrapping table · filing cabinet

CAN BE STOOD ON

stool · rug

Notes

WHO	POSSIBLE SQUARES
JF Jack Frost	
AS Alabaster Snowball	
SA Santa	
JL Jewel Loggins	
T target	

Code grid

	A	B	C	D	E
1	8	f	Y	C	R
2	8	S	t	R	t
3	r	n	N	F	G
4	N	9	Q	n	n
5	R	b	H	a	n

The character on the culprit's square is character **3** of the password.

Culprit _____ **No. 3** ☐

CASE 04

LEVEL 1 · RECRUIT

The Master Key

There is exactly one master key for this floor. It opens every door, every cabinet and, for reasons nobody can explain, the cookie safe. It lives in a locked box with a sign-out sheet. Today it was found lying about in the open with a little red bow on it, as if it were a present. Morcel Nougat nearly slipped on it. The sign-out sheet is blank, the box is open, and everybody has suddenly become very interested in their own shoes.

TARGET
the master key

Suspects and clues

Jack Frost (JF)

1. Jack Frost was right next to a fireplace.

Morcel Nougat (MN)

2. Morcel Nougat was on a beanbag.

Jingle Ringford (JR)

3. Jingle Ringford was by the north outer wall.
4. Jingle Ringford was in the east half of the floor.

Santa (SA)

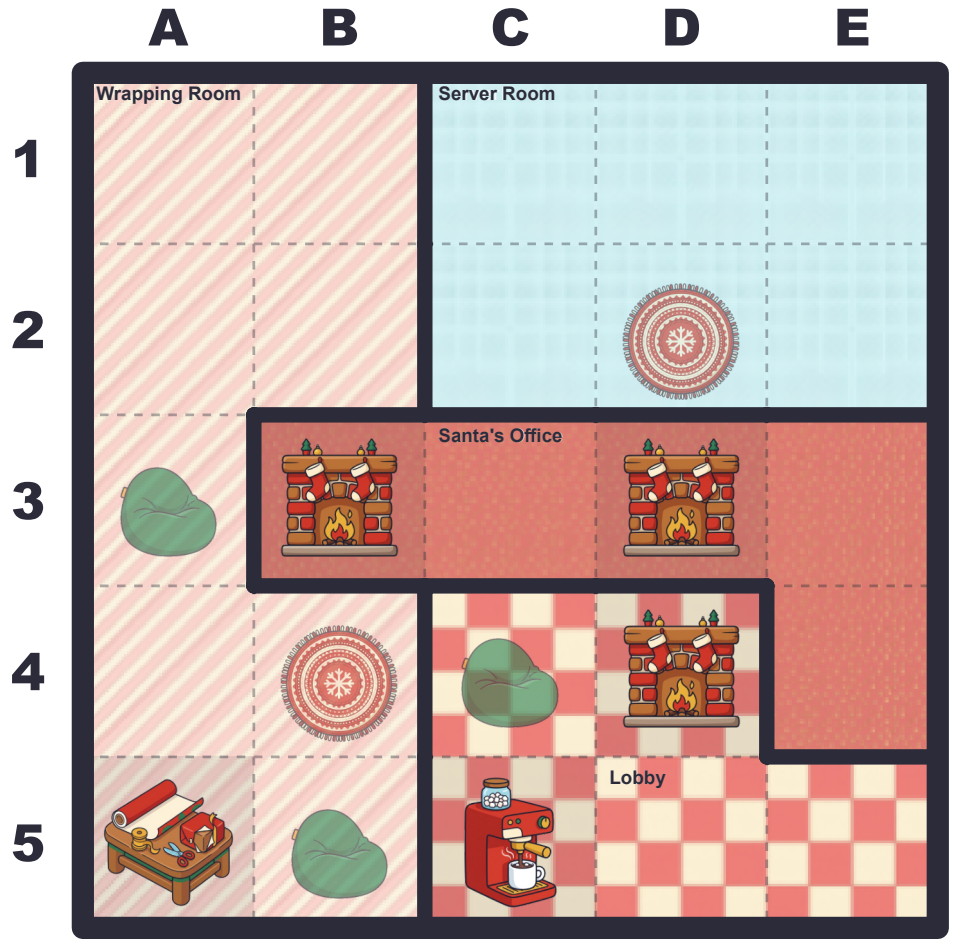
5. Santa was in the Wrapping Room.
6. Santa was on a rug.

CASE 04

The Master Key

Map of the floor · 5 × 5 · north is up

N



ROOMS

Santa's Office · Server Room · Wrapping Room · Lobby

BLOCKED: NOBODY STANDS HERE

fireplace · wrapping table · cocoa machine

CAN BE STOOD ON

rug · beanbag

Notes

WHO	POSSIBLE SQUARES
JF Jack Frost	
MN Morcel Nougat	
JR Jingle Ringford	
SA Santa	
T target	

Code grid

	A	B	C	D	E
1	C	r	P	a	L
2	V	3	f	Y	T
3	y	9	g	n	K
4	e	B	L	n	y
5	h	j	C	L	L

The character on the culprit's square is character **4** of the password.

Culprit _____ **No. 4** ☐

CASE 05

LEVEL 1 · RECRUIT

The Open Folder

The folder with this year's delivery routes is stamped 'TOP SECRET' on the front, on the back and, for the avoidance of doubt, on the spine. At the end of every shift it goes into a locked cabinet. Last night it went nowhere. It was found this morning lying wide open, pages fanned out like a buffet, with a cocoa ring on the map of the southern hemisphere. Somebody clocked out in a hurry. Somebody will shortly be explaining the words 'clean desk policy' to Santa.

TARGET
the route folder

Suspects and clues

Jack Frost (JF)

1. Jack Frost was in the south half of the floor.
2. Jack Frost was in a room with a printer.

Santa (SA)

3. Santa was in a room with a chair.

Pepper Minstix (PM)

4. Pepper Minstix was on a rug.

Ribb Bonbowford (RB)

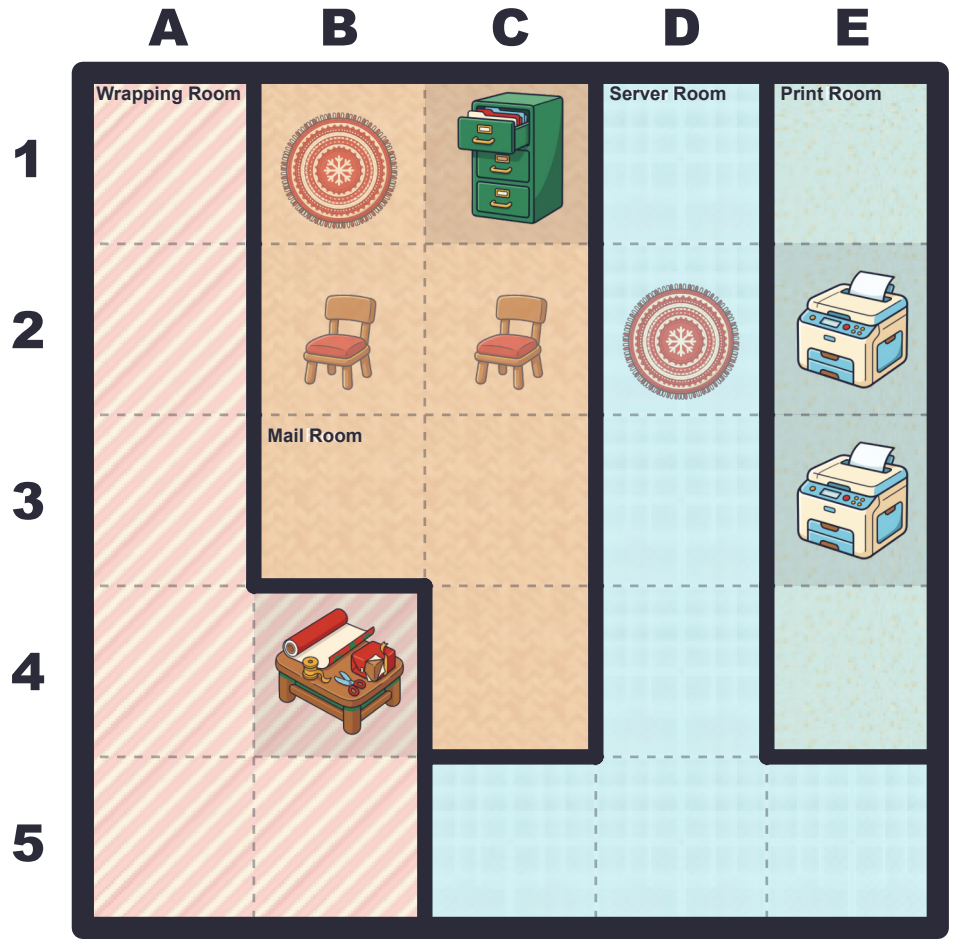
5. Ribb Bonbowford was right next to a wrapping table.

CASE 05

The Open Folder

Map of the floor · 5 × 5 · north is up

N



ROOMS

Mail Room · Wrapping Room · Print Room · Server Room

BLOCKED: NOBODY STANDS HERE

filing cabinet · printer · wrapping table

CAN BE STOOD ON

rug · chair

Notes

WHO	POSSIBLE SQUARES
JF Jack Frost	
SA Santa	
PM Pepper Minstix	
RB Ribb Bonbowford	
T target	

Code grid

	A	B	C	D	E
1	9	6	Y	V	T
2	U	L	N	R	3
3	A	a	m	8	K
4	Q	Q	j	2	e
5	m	m	X	J	L

The character on the culprit's square is character **5** of the password.

Culprit _____ **No. 5** ☐

CASE 06

LEVEL 1 · RECRUIT

The Visitor Badge

Visitors sign in at the front desk, show some identification and receive a badge. It is not a complicated system; a reasonably bright snowman could run it. And yet this morning a visitor badge was active and strolling around the building while the visitor book showed no entry at all. No name, no signature, not even a doodle. Somebody did a guest a favor and skipped the paperwork. The guest, it turns out, had not come for the tour.

TARGET
the visitor badge

Suspects and clues

Wombley Cube (WC)

1. Wombley Cube was right next to a chair.

Santa (SA)

2. Santa was in the Mail Room.

Jack Frost (JF)

3. Jack Frost was right next to a parcel shelf.

Morcel Nougat (MN)

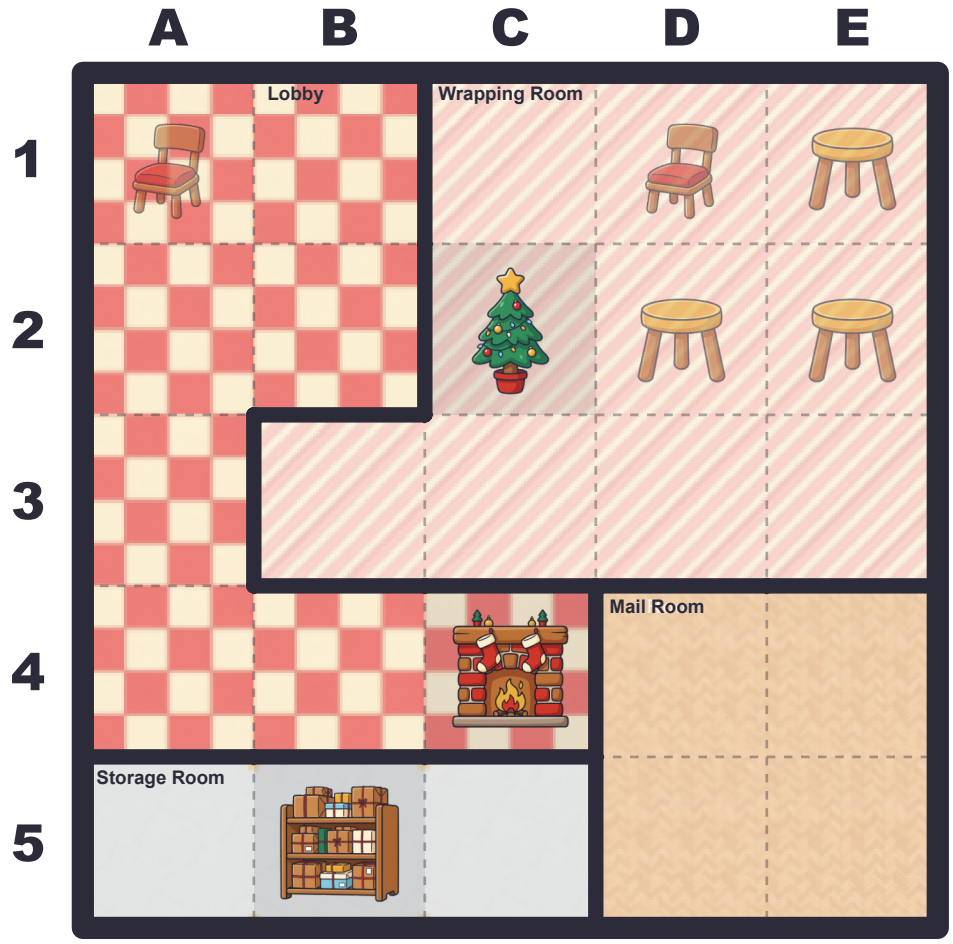
4. Morcel Nougat was by the west outer wall.
5. Morcel Nougat was by the north outer wall.

CASE 06

The Visitor Badge

Map of the floor · 5 × 5 · north is up

N



ROOMS

Wrapping Room · Storage Room · Lobby · Mail Room

BLOCKED: NOBODY STANDS HERE

Christmas tree · fireplace · parcel shelf

CAN BE STOOD ON

chair · stool

Notes

WHO — POSSIBLE SQUARES

WC Wombley Cube

SA Santa

JF Jack Frost

MN Morcel Nougat

T target

Code grid

	A	B	C	D	E
1	e	5	h	G	K
2	K	5	D	K	D
3	C	t	d	K	m
4	3	d	A	7	K
5	Z	U	6	t	9

The character on the culprit's square is character **6** of the password.

Culprit _____ **No. 6** ☐

THE STORY SO FAR

Chapter 2: I Don't Recall

Six cases, six culprits, six characters of the password. Santa is delighted and promotes you to Responder, a title that comes with a lanyard and no extra pay.

You have noticed a pattern. Nobody meant any harm. Each of them was simply being helpful, and each remembers a friendly stranger who made a suggestion just beforehand: blue scarf, cold hands, very polite. Nobody remembers a face.

Word of your success has spread, and with it a sudden outbreak of poor memory. Witnesses who last week could name the exact chair they sat on now 'think they were somewhere near the middle, possibly'. Some have nothing to say at all. You will have to make do with less.

LEVEL 2

Responder

CASES	MAP	SUSPECTS	NEW
07 - 12	5 × 5	4	Ask the rows and columns

How to solve this level

The clues get thinner. Some people have only a vague clue, some have none, and nobody may be down to one square at the start. When you are stuck, stop asking where a person can be and ask who can stand in a certain row or column. Every row and every column needs exactly one figure. If only one figure can still reach a row, that figure is in that row, however many other squares it seemed to have. Clues can now also say where somebody was not. Guide: chapter 7.

The vault password so far

Write down the character from every solved case. All thirty have to be right.

LEVEL 1						LEVEL 2						LEVEL 3						LEVEL 4						LEVEL 5					
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30

CASE 07

LEVEL 2 · RESPONDER

The Borrowed Badge

Every elf has a staff badge, and every badge opens exactly the doors its owner needs. This sensible arrangement lasted until somebody forgot theirs at home and a colleague spoke the five most expensive words in security: 'Here, just take mine.' The borrowed badge has since opened four doors its real owner never uses, spent an hour on a windowsill and come back with a thank-you biscuit. The door system now believes one elf was in two places at once. Candycane Striper finds this hilarious. Santa does not.

TARGET
the staff badge

Suspects and clues

Santa — SA

1. Santa was in a room with a fireplace.

Candycane Striper — CC

2. Candycane Striper was on a chair.

Jack Frost — JF

3. Jack Frost was in the north half of the floor.

Jewel Loggins — JL

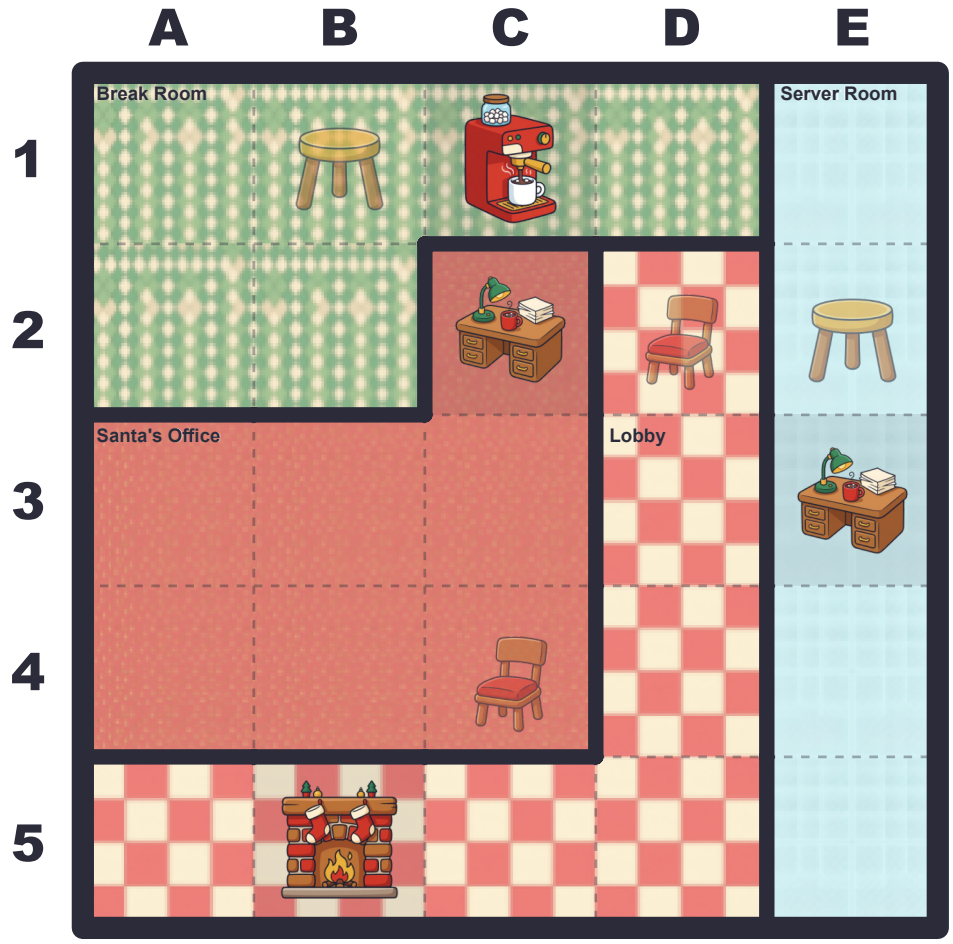
4. Jewel Loggins was right next to a cocoa machine.
5. Jewel Loggins was in the east half of the floor.

CASE 07

The Borrowed Badge

Map of the floor · 5 × 5 · north is up

N



ROOMS

Santa's Office · Server Room · Lobby · Break Room

BLOCKED: NOBODY STANDS HERE

cocoa machine · desk · fireplace

CAN BE STOOD ON

stool · chair

Notes

WHO	POSSIBLE SQUARES
SA Santa	
CC Candycane Striper	
JF Jack Frost	
JL Jewel Loggins	
T target	

Code grid

	A	B	C	D	E
1	6	W	8	U	6
2	F	C	S	E	j
3	A	T	6	J	4
4	A	C	6	L	m
5	K	T	E	V	V

The character on the culprit's square is character **7** of the password.

Culprit _____ **No. 7** ☐

CASE 08

LEVEL 2 · RESPONDER

The Forgotten Printout

Somebody printed the complete list of this year's sleigh stopovers, all forty-one pages of it, and then went to lunch. The printout sat in the open for two hours, face up, on a floor where half the staff walk past on their way to the cocoa. When its owner finally remembered it, the pages were still there, but in a different order, and page seventeen had a thumbprint in icing sugar. Nobody saw anything. Several people have suddenly lost their appetite.

TARGET
the printout

Suspects and clues

Morcel Nougat (MN)

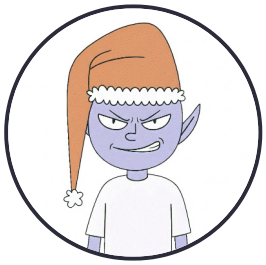
1. Morcel Nougat was in the west half of the floor.
2. Morcel Nougat was in a room with a workbench.

Jack Frost (JF)

3. Jack Frost was in a room with a cocoa machine.

Santa (SA)

4. Santa was right next to a stool.
5. Santa was in the east half of the floor.



Poinsettia Mcmittens (PO)

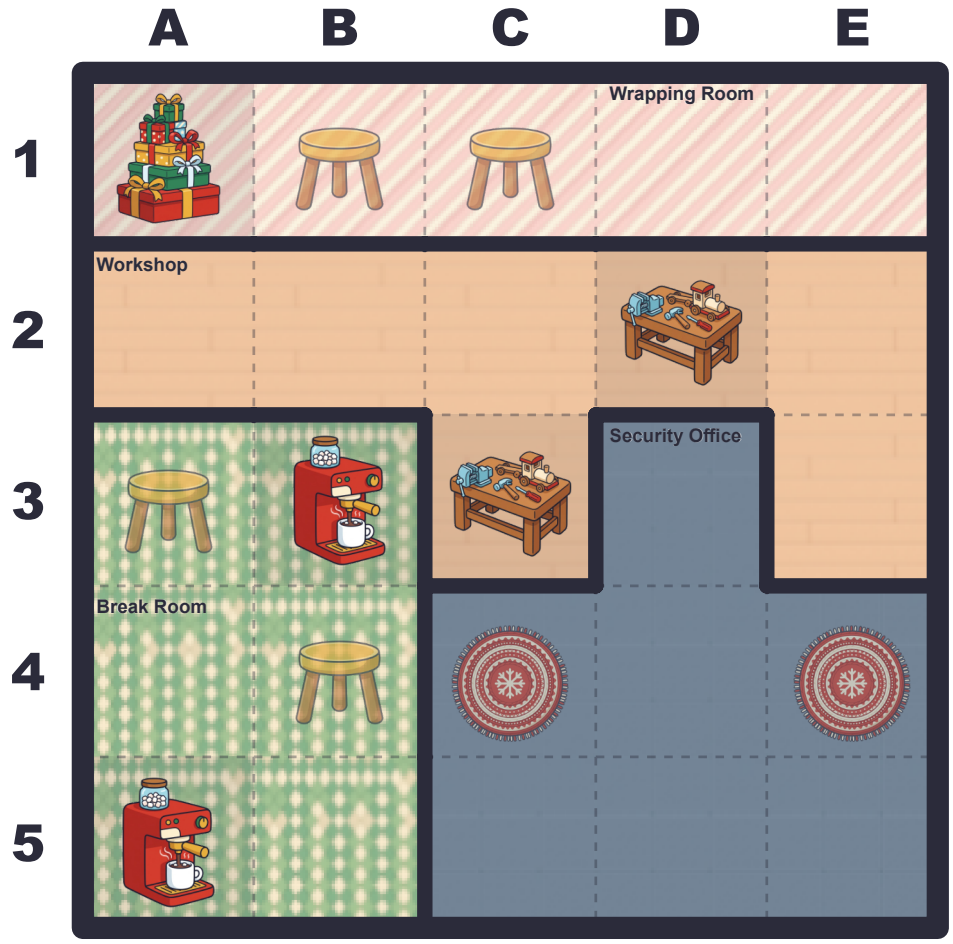
6. Poinsettia Mcmittens was by the south outer wall.
7. Poinsettia Mcmittens was in the east half of the floor.

CASE 08

The Forgotten Printout

Map of the floor · 5 × 5 · north is up

N



ROOMS

Wrapping Room · Break Room · Security Office · Workshop

BLOCKED: NOBODY STANDS HERE

gift pile · workbench · cocoa machine

CAN BE STOOD ON

stool · rug

Notes

WHO	POSSIBLE SQUARES
MN Morcel Nougat	
JF Jack Frost	
SA Santa	
PO Poinsettia Mcmittens	
T target	

Code grid

	A	B	C	D	E
1	a	9	J	D	4
2	Q	J	3	L	Y
3	b	8	t	y	G
4	Q	q	4	2	V
5	d	e	W	f	Q

The character on the culprit's square is character **8** of the password.

Culprit _____ **No. 8** ☐

CASE 09

LEVEL 2 · RESPONDER

Remind Me Tomorrow

The workshop computer has been asking for the same security update since the spring. Every morning a small window appears, and every morning somebody clicks 'Remind me tomorrow', a button that has now been pressed one hundred and ninety-three times. Yesterday the reminders stopped. This was not good news: something else had finally installed itself instead. The computer now plays sleigh bells at random and sends its files somewhere cold. Somebody clicked that button for the last time.

TARGET
the workshop computer

Suspects and clues

Santa (SA)

1. Santa was right next to a gift pile.
2. Santa was in a room with a wrapping table.

Jack Frost (JF)

3. Jack Frost was by the west outer wall.

Ginger Breddie (GB)

4. Ginger Breddie was on a beanbag.

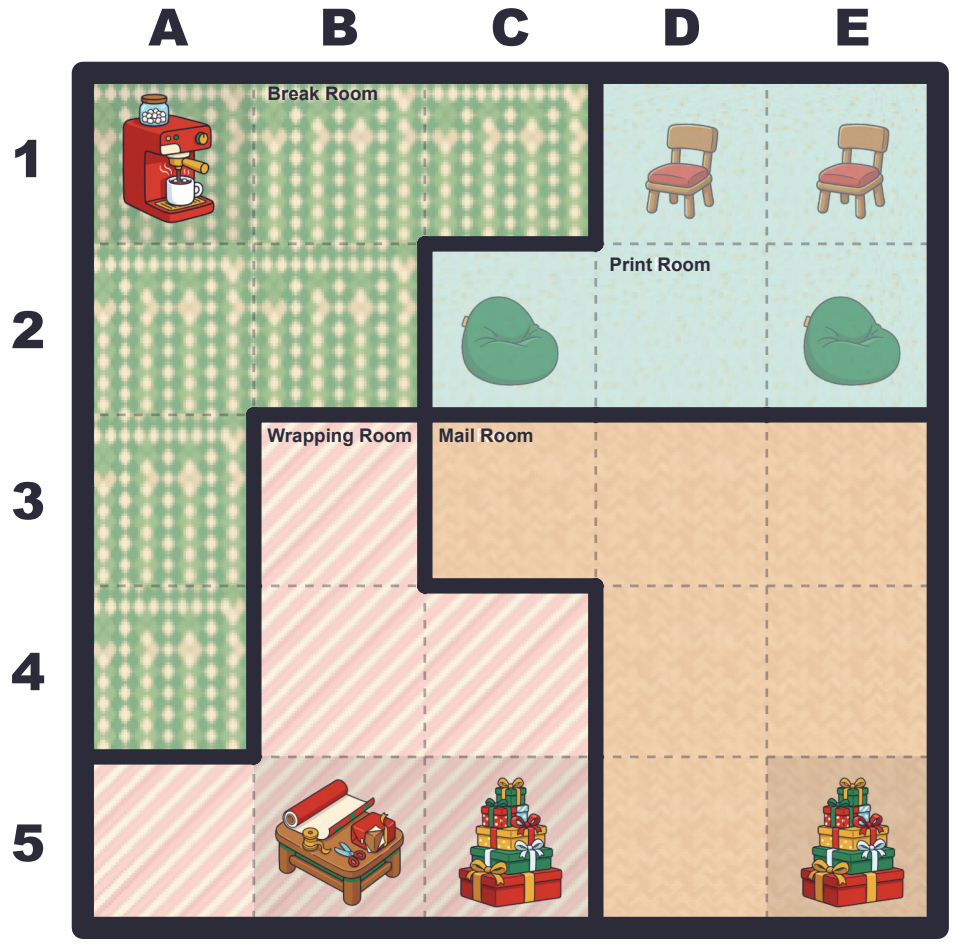
Garland Candlesticks (GC)

5. Garland Candlesticks was by the south outer wall.

Remind Me Tomorrow

Map of the floor · 5 × 5 · north is up

N



ROOMS

Print Room · Wrapping Room · Mail Room · Break Room

BLOCKED: NOBODY STANDS HERE

cocoa machine · wrapping table · gift pile

CAN BE STOOD ON

chair · beanbag

Notes

WHO	POSSIBLE SQUARES
SA Santa	
JF Jack Frost	
GB Ginger Breddie	
GC Garland Candlesticks	
T target	

Code grid

	A	B	C	D	E
1	U	m	t	J	9
2	G	7	5	C	3
3	J	G	A	C	J
4	3	h	D	D	C
5	i	S	a	q	X

The character on the culprit's square is character **9** of the password.

Culprit _____ **No. 9** ☐

CASE 10

LEVEL 2 · RESPONDER

The Lost Phone

A work phone was found yesterday, face up and unlocked, with the mail app open. It has no screen lock, because its owner considers a six-digit code 'a bit much before breakfast'. The phone had been lying there long enough for somebody to scroll back through three weeks of messages and forward the staff holiday rota to an outside address. The owner has been identified and is very sorry. What we still need is the person who picked it up.

TARGET
the work phone

Suspects and clues

Angel Candysalt (AC)

1. Angel Candysalt was in the north half of the floor.
2. Angel Candysalt was in a room with a chair.

Jack Frost (JF)

3. Jack Frost was by the north outer wall.

Santa (SA)

4. Santa was by the east outer wall.
5. Santa was not in the Wrapping Room.

Ginger Breddie (GB)

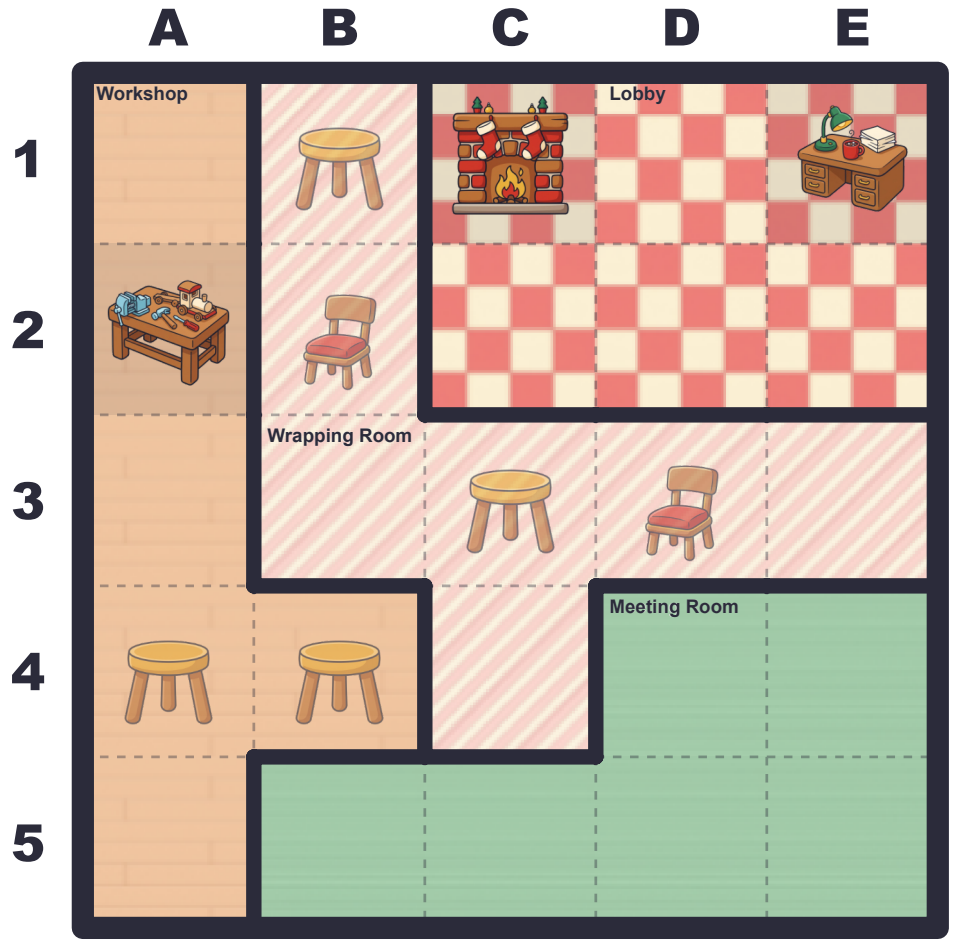
6. Ginger Breddie was by the west outer wall.
7. Ginger Breddie was on a stool.

CASE 10

The Lost Phone

Map of the floor · 5 × 5 · north is up

N



ROOMS

Lobby · Meeting Room · Wrapping Room · Workshop

BLOCKED: NOBODY STANDS HERE

fireplace · desk · workbench

CAN BE STOOD ON

stool · chair

Notes

WHO — POSSIBLE SQUARES

AC Angel Candysalt

JF Jack Frost

SA Santa

GB Ginger Breddie

T target

Code grid

	A	B	C	D	E
1	D	q	P	g	a
2	a	g	H	A	g
3	L	E	P	W	5
4	N	6	J	X	d
5	g	2	q	t	R

The character on the culprit's square is character **10** of the password.

Culprit _____ **No. 10** ☐

CASE 11

LEVEL 2 · RESPONDER

The Free Font Pack

The gift tags needed a more festive typeface. Instead of asking IT, somebody searched for 'free christmas fonts no virus', a search that tends to deliver the opposite. The font pack installed itself together with a toolbar, a weather widget for a town nobody has heard of, and a program that takes a screenshot every thirty seconds. The gift tags do look lovely. The design computer has been unplugged and is sitting in a corner, thinking about what it has done.

TARGET
the design computer

Suspects and clues

Eve Snowshoes (ES)

1. Eve Snowshoes was in the north half of the floor.
2. Eve Snowshoes was in a room with a gift pile.

Bow Ninecandle (BN)

3. Bow Ninecandle was right next to a printer.

Jack Frost (JF)

4. Jack Frost was by the west outer wall.
5. Jack Frost was in the Archive.

Santa (SA)

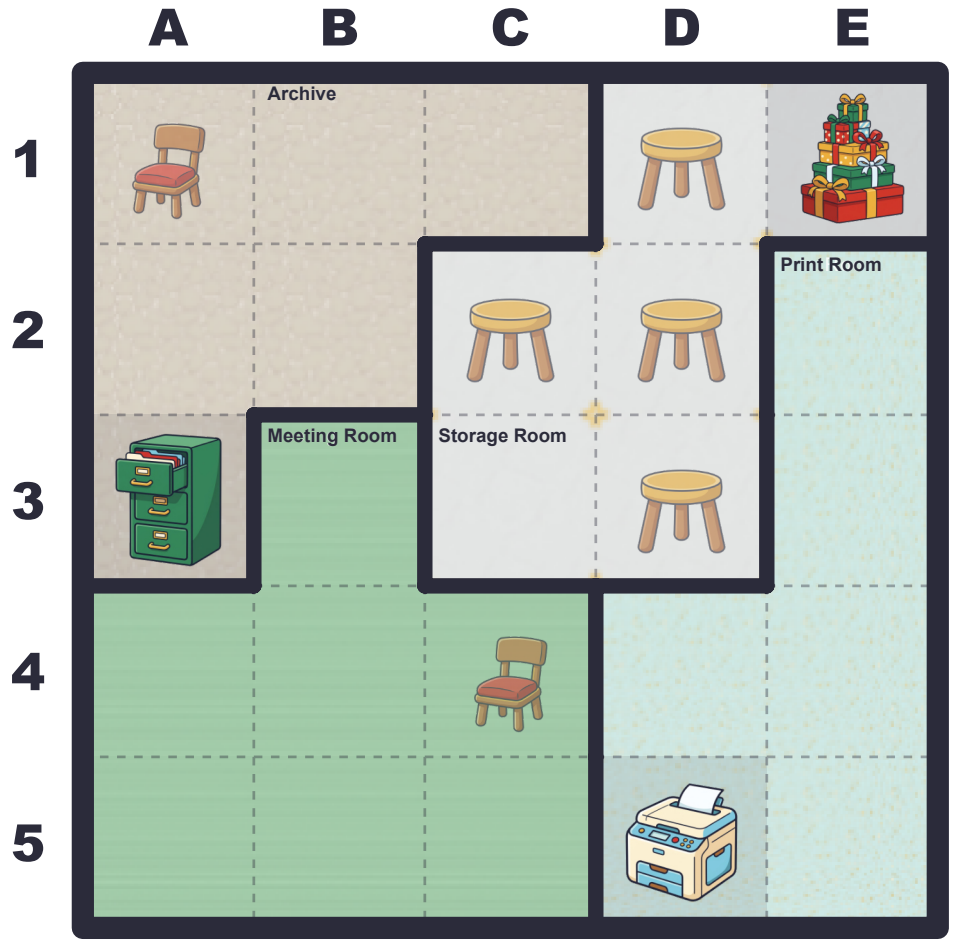
6. Santa was by the south outer wall.
7. Santa was in the west half of the floor.

CASE 11

The Free Font Pack

Map of the floor · 5 × 5 · north is up

N



ROOMS

Storage Room · Archive · Print Room · Meeting Room

BLOCKED: NOBODY STANDS HERE

gift pile · filing cabinet · printer

CAN BE STOOD ON

chair · stool

Notes

WHO — POSSIBLE SQUARES

ES Eve Snowshoes

BN Bow Ninecandle

JF Jack Frost

SA Santa

T target

Code grid

	A	B	C	D	E
1	e	a	q	W	r
2	9	H	r	t	Q
3	y	d	P	J	A
4	2	T	8	2	B
5	n	2	R	n	9

The character on the culprit's square is character **11** of the password.

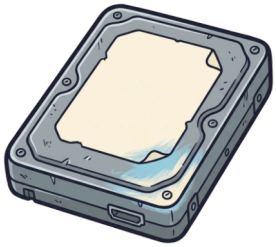
Culprit _____ **No. 11** ☐

CASE 12

LEVEL 2 · RESPONDER

The Recycling Bin

The old gift database was retired last month, and its hard drive was to be wiped and shredded. Instead it was found in the paper recycling, between a pizza box and last year's calendar, with every file intact and a label reading 'OLD - PROBABLY EMPTY'. It is not empty. It holds eleven years of addresses. Somebody carried it to the bin, and somebody else has had it out again since: there are fresh fingerprints on the casing, and they are cold.



TARGET
the old hard drive

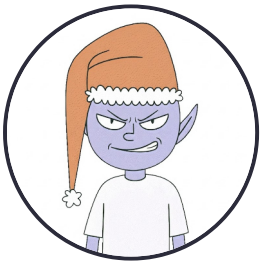
Suspects and clues

Jack Frost (JF)

1. Jack Frost was by the north outer wall.
2. Jack Frost was in a room with a cocoa machine.

Santa (SA)

3. Santa was right next to a cocoa machine.



Poinsettia Mcmittens (PO)

4. Poinsettia Mcmittens was on a beanbag.
5. Poinsettia Mcmittens was in a room with a filing cabinet.

Angel Candysalt (AC)

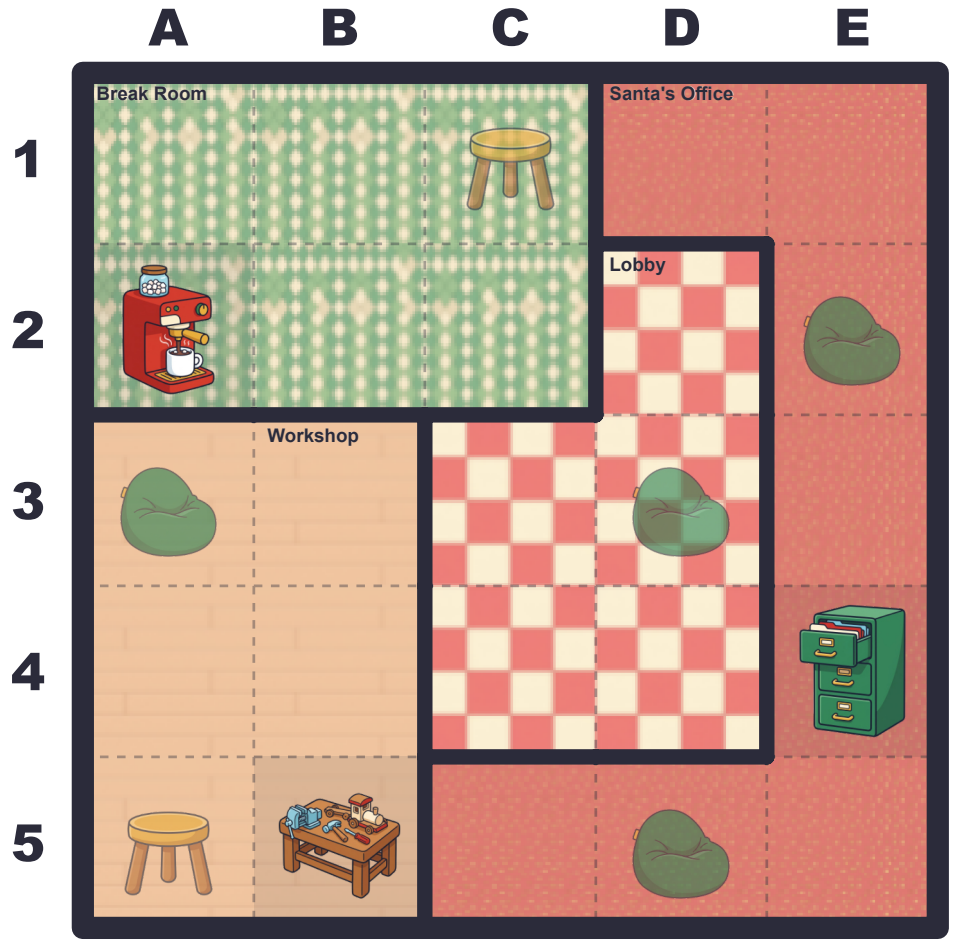
6. Angel Candysalt was in the south half of the floor.
7. Angel Candysalt was not in the Lobby.

CASE 12

The Recycling Bin

Map of the floor · 5 × 5 · north is up

N



ROOMS

Workshop · Break Room · Lobby · Santa's Office

BLOCKED: NOBODY STANDS HERE

cocoa machine · filing cabinet · workbench

CAN BE STOOD ON

stool · beanbag

Notes

WHO	POSSIBLE SQUARES
JF Jack Frost	
SA Santa	
PO Poinsettia Mcmittens	
AC Angel Candysalt	
T target	

Code grid

	A	B	C	D	E
1	M	R	b	G	n
2	W	h	G	5	6
3	e	H	q	y	C
4	6	R	a	y	4
5	J	E	3	R	f

The character on the culprit's square is character **12** of the password.

Culprit _____ **No. 12** ☐

THE STORY SO FAR

Chapter 3: Compared to What

Twelve characters. The vault door has started to look less smug.

The elves, meanwhile, have discovered a new way of being unhelpful. Nobody wants to say which room they were in, in case it turns out to be the wrong one. So they describe themselves in terms of each other: 'I was further north than Pepper.' 'I was in the same room as Jingle, wherever that was.' It is like taking witness statements from a flock of birds.

Santa promotes you to Analyst. And one name has now been on every single suspect list: Jack Frost, visiting consultant for 'seasonal optimization'. He is charming, he is cooperative, and he is always around. So far he has been guilty only some of the time, which is somehow worse.

LEVEL 3

Analyst

CASES	MAP	SUSPECTS	NEW
13 - 18	6 × 6	5	Comparing clues

How to solve this level

The map grows to six by six with five people, and most clues now compare two figures: further north than, in the same room as, the only person in the room. One such clue rarely places anybody. Turn each one into limits (X cannot be in the bottom row, Y cannot be in the top row) and let two or three clues squeeze the same person. The target may now appear in a comparison, but never in a clue about its room. Guide: chapter 8.

The vault password so far

Write down the character from every solved case. All thirty have to be right.

LEVEL 1						LEVEL 2						LEVEL 3						LEVEL 4						LEVEL 5					
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30

CASE 13

LEVEL 3 · ANALYST

A Parcel from the Easter Bunny

At 9:02 an e-mail reached every inbox on the floor: 'Your parcel from the Easter Bunny could not be delivered. Click here to rearrange.' It is December. The Easter Bunny has never sent anything to this address, the sender's domain was spelled 'nroth-pole', and the message opened with 'Dear valued elf or current resident'. By 9:03 somebody had clicked the link and typed in their password. They would like it noted that they really were expecting a parcel.

TARGET
the mail terminal

Suspects and clues

Santa (SA)

No clue of their own. Look at the others.

Wombley Cube (WC)

1. Wombley Cube was right next to a cocoa machine.
2. Wombley Cube was further east than Santa.
3. Wombley Cube was in the north half of the floor.

Alabaster Snowball (AS)

4. Alabaster Snowball was not in the same room as Jewel Loggins.

Jewel Loggins (JL)

5. Jewel Loggins was right next to a chair.

Jack Frost (JF)

6. Jack Frost was right next to a rug.
7. Jack Frost was by the east outer wall.
8. Jack Frost was further south than the mail terminal.

More clues

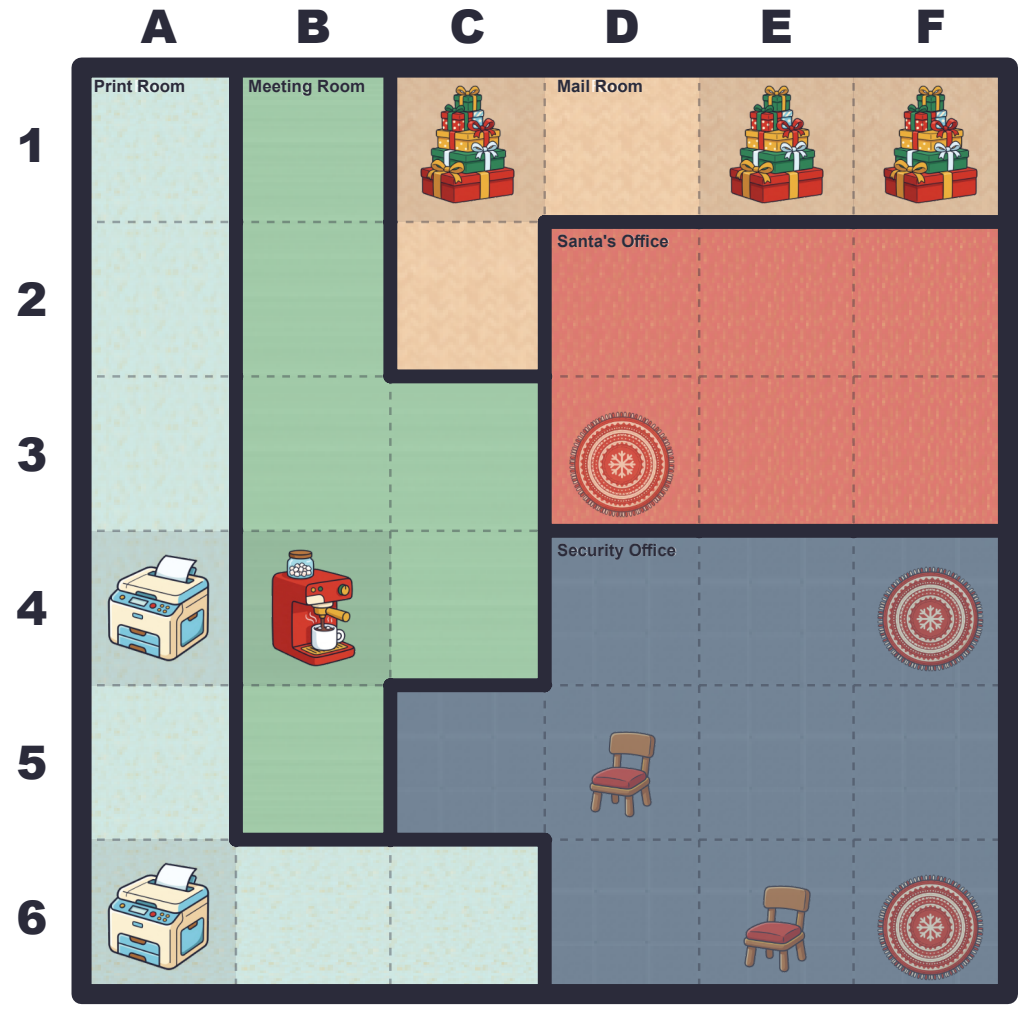
9. The mail terminal was further west than Jewel Loggins.

CASE 13

A Parcel from the Easter Bunny

Map of the floor · 6 × 6 · north is up

N



ROOMS

Print Room · Mail Room · Security Office · Meeting Room · Santa's Office

BLOCKED: NOBODY STANDS HERE

gift pile · printer · cocoa machine

CAN BE STOOD ON

rug · chair

Notes

WHO	POSSIBLE SQUARES
SA Santa	
WC Wombley Cube	
AS Alabaster Snowball	
JL Jewel Loggins	
JF Jack Frost	
T target	

Code grid

	A	B	C	D	E	F
1	b	7	F	P	P	d
2	X	d	2	f	A	e
3	R	D	e	A	F	m
4	H	R	S	g	3	D
5	K	P	4	U	N	q
6	n	D	L	D	h	N

The character on the culprit's square is character **13** of the password.

Culprit _____ **No. 13** ☐

CASE 14

LEVEL 3 · ANALYST

A Voice Like Santa's

The desk phone rang in the middle of the afternoon cocoa break. A warm, familiar voice said it was Santa, that he was locked out, and that he needed the alarm code 'right now, ho ho ho'. The real Santa was in the building at the time, in plain sight, eating a gingerbread man. He has also never once said 'ho ho ho' on the telephone, because he considers it unprofessional. Somebody read out the code anyway, digit by digit, and wished the caller a lovely day.

TARGET
the desk phone

Suspects and clues

Noel Boetie (NB)

1. Noel Boetie was in a room with a cocoa machine.
2. Noel Boetie was further west than Jack Frost.

Ginger Breddie (GB)

3. Ginger Breddie was the only person in the room.
4. Ginger Breddie was further west than Chimney Scissorsticks.

Santa (SA)

5. Santa was right next to a beanbag.
6. Santa was by the east outer wall.

Jack Frost (JF)

No clue of their own. Look at the others.

Chimney Scissorsticks (CS)

7. Chimney Scissorsticks was right next to a desk.
8. Chimney Scissorsticks was in a room with a chair.
9. Chimney Scissorsticks was further south than Noel Boetie.

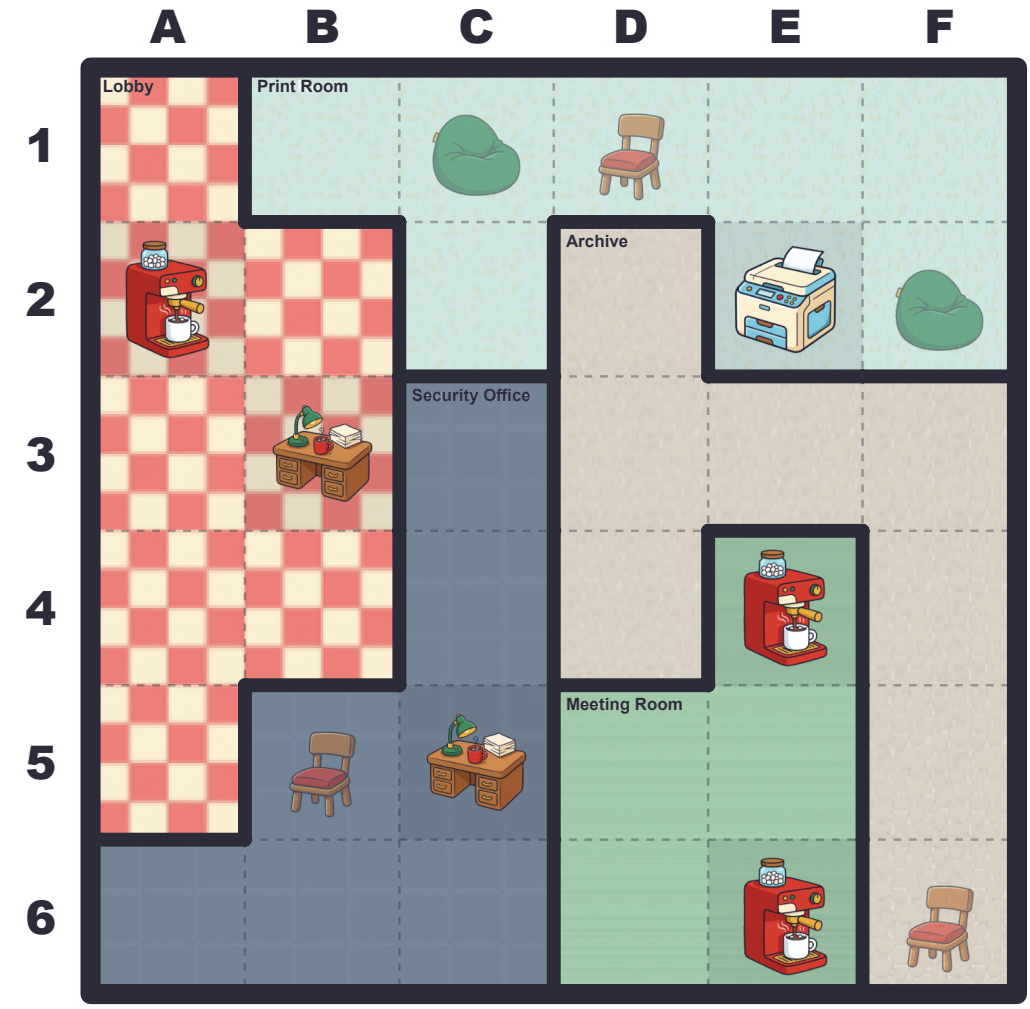
More clues

10. The desk phone was further north than Jack Frost.

A Voice Like Santa's

Map of the floor · 6 × 6 · north is up

N



ROOMS

Lobby · Meeting Room · Print Room · Archive · Security Office

BLOCKED: NOBODY STANDS HERE

cocoa machine · printer · desk

CAN BE STOOD ON

beanbag · chair

Notes

WHO	POSSIBLE SQUARES
NB Noel Boetie	
GB Ginger Breddie	
SA Santa	
JF Jack Frost	
CS Chimney Scissorsticks	
T target	

Code grid

	A	B	C	D	E	F
1	U	H	3	9	8	G
2	R	m	n	P	R	r
3	b	g	3	3	T	W
4	3	6	8	e	6	L
5	K	W	n	D	9	h
6	R	P	h	i	S	Q

The character on the culprit's square is character **14** of the password.

Culprit _____ **No. 14** ☐

CASE 15

LEVEL 3 · ANALYST

After You!

The door has a badge reader, a camera, a warning sign and a second, larger warning sign. Yesterday a visitor arrived behind a tower of gift boxes so tall that only his shoes were showing. He had no badge. He did not need one, because somebody on the inside saw a person struggling with parcels and did the decent thing. Kindness is the finest quality an elf can have. It is also, as the security team points out rather wearily, the easiest one to exploit.

TARGET
the badge reader

Suspects and clues

Jack Frost (JF)

1. Jack Frost was right next to a beanbag.
2. Jack Frost was further east than Pepper Minstix.

Alabaster Snowball (AS)

No clue of their own. Look at the others.

Pepper Minstix (PM)

3. Pepper Minstix was the only person in the room.
4. Pepper Minstix was further east than Alabaster Snowball.

Santa (SA)

5. Santa was not in the Security Office.

Noel Boetie (NB)

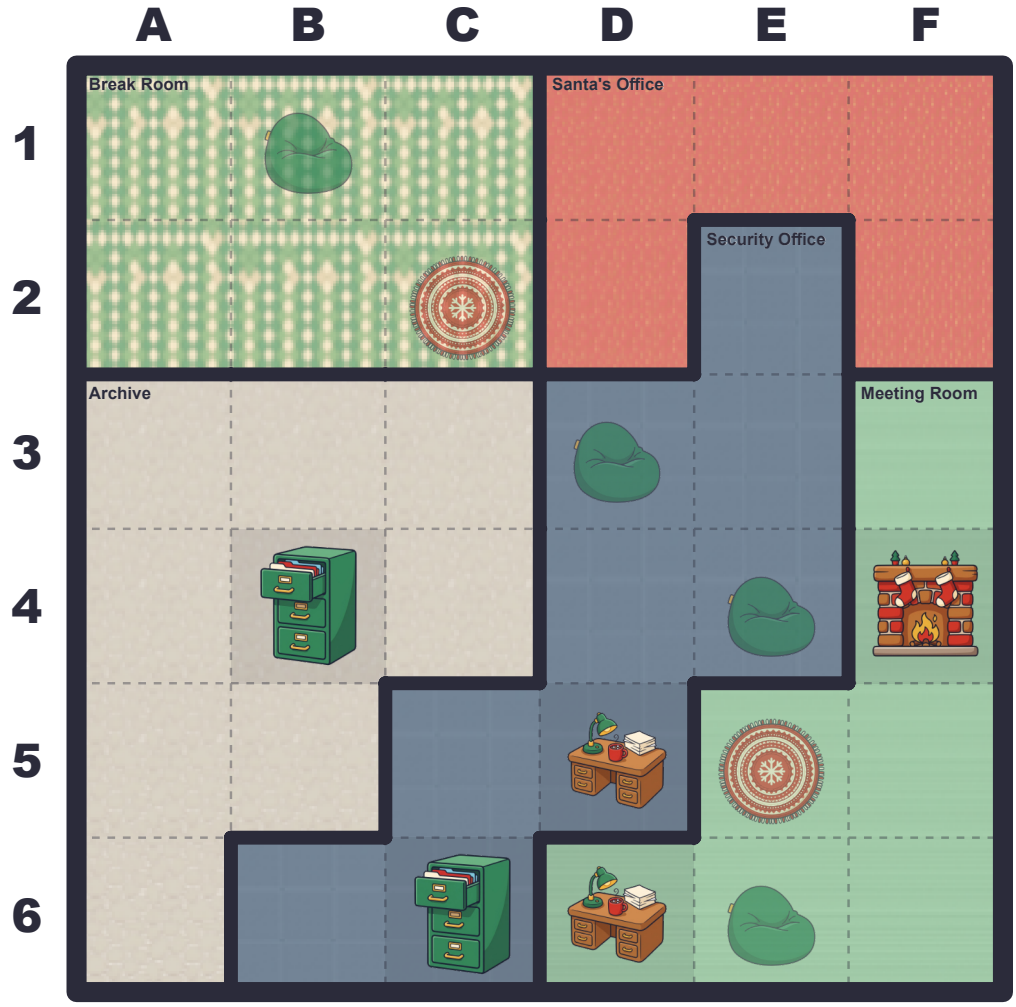
6. Noel Boetie was in the south half of the floor.
7. Noel Boetie was in the Security Office.
8. Noel Boetie was on a beanbag.
9. Noel Boetie was further south than Jack Frost.

CASE 15

After You!

Map of the floor · 6 × 6 · north is up

N



ROOMS

Meeting Room · Archive · Break Room · Santa's Office · Security Office

BLOCKED: NOBODY STANDS HERE

filing cabinet · fireplace · desk

CAN BE STOOD ON

beanbag · rug

Notes

WHO	POSSIBLE SQUARES
JF Jack Frost	
AS Alabaster Snowball	
PM Pepper Minstix	
SA Santa	
NB Noel Boetie	
T target	

Code grid

	A	B	C	D	E	F
1	M	E	i	M	S	3
2	7	F	V	9	K	G
3	y	d	f	L	f	Q
4	d	3	g	Z	M	f
5	L	G	m	3	d	m
6	R	m	V	a	M	9

The character on the culprit's square is character **15** of the password.

Culprit _____ **No. 15**

CASE 16

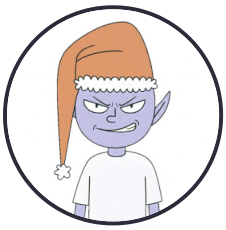
LEVEL 3 · ANALYST

The Confident Chatbot

The floor has a new chatbot, and this year somebody let it write the security report. It took four seconds and produced eleven pages. The report praises a firewall brand that does not exist, lists an open port numbered 88555 and describes last year's audit, which never took place, as 'a resounding success'. All of it in flawless, confident prose. It was forwarded to Santa unread. Poinsettia Mcmittens found the port number in the afternoon and has been staring at a wall ever since.

TARGET
the chatbot terminal

Suspects and clues



Poinsettia Mcmittens

PO

No clue of their own. Look at the others.

Santa

SA

1. Santa was right next to a stool.
2. Santa was further south than Bow Ninecandle.
3. Santa was further west than Poinsettia Mcmittens.

Jack Frost

JF

4. Jack Frost was in the north half of the floor.
5. Jack Frost was in a room with a rug.
6. Jack Frost was in a room with a fireplace.
7. Jack Frost was further south than Jingle Ringford.
8. Jack Frost was further south than Poinsettia Mcmittens.

Bow Ninecandle

BN

9. Bow Ninecandle was in the Break Room.

Jingle Ringford

JR

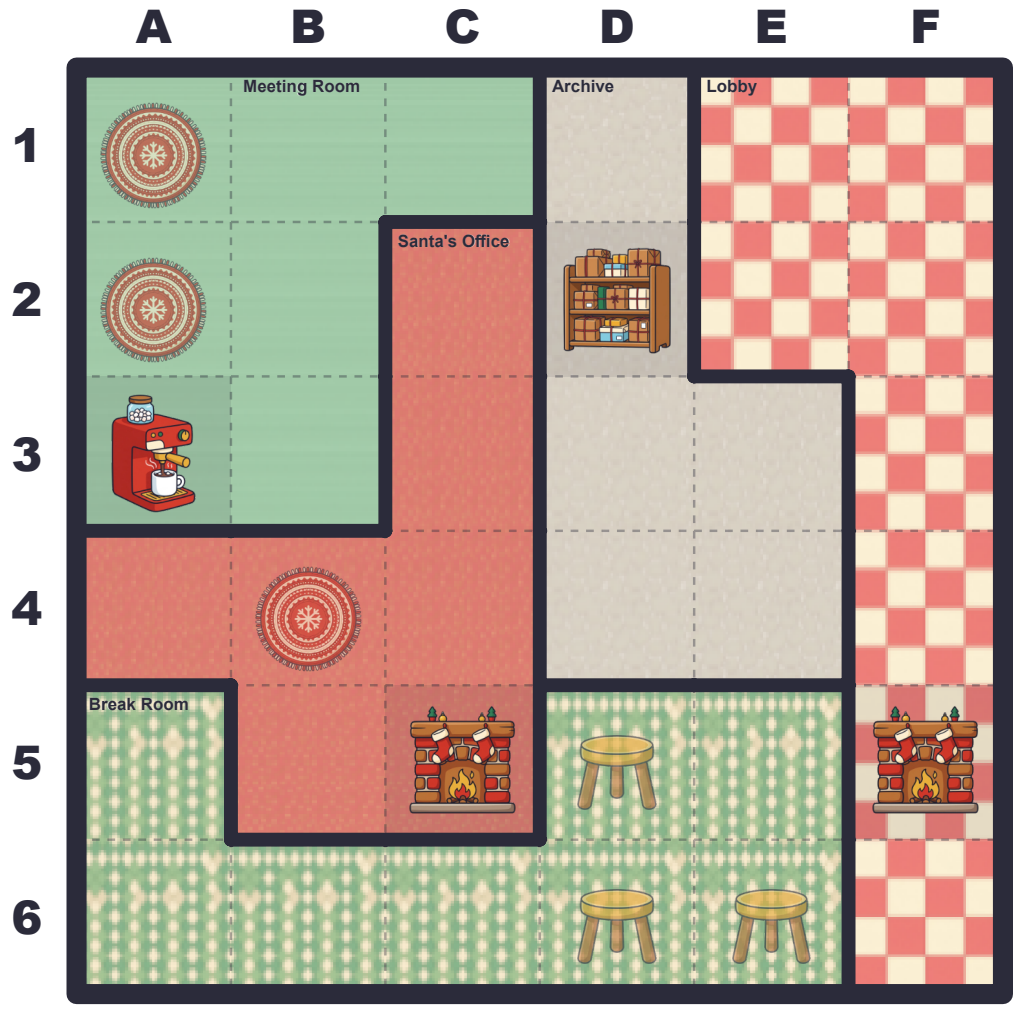
10. Jingle Ringford was in a room with a parcel shelf.

CASE 16

The Confident Chatbot

Map of the floor · 6 × 6 · north is up

N



ROOMS

Santa's Office · Lobby · Archive · Break Room · Meeting Room

BLOCKED: NOBODY STANDS HERE

parcel shelf · cocoa machine · fireplace

CAN BE STOOD ON

rug · stool

Notes

WHO	POSSIBLE SQUARES
PO Poinsettia Mcmittens	
SA Santa	
JF Jack Frost	
BN Bow Ninecandle	
JR Jingle Ringford	
T target	

Code grid

	A	B	C	D	E	F
1	7	d	h	d	7	3
2	5	9	r	Z	2	b
3	f	6	5	t	G	7
4	3	F	N	N	W	5
5	e	a	B	N	D	A
6	2	C	Y	T	R	h

The character on the culprit's square is character **16** of the password.

Culprit _____ **No. 16** ☐

CASE 17

LEVEL 3 · ANALYST

The Group Photo

It is a lovely photo. Everyone is smiling, the tree is lit, and somebody is doing rabbit ears behind Santa. It went online at lunchtime and collected four hundred likes within the hour. Unfortunately the photographer picked the spot in front of the whiteboard, and the whiteboard, perfectly in focus, shows this week's alarm codes under the heading 'ALARM CODES - DO NOT SHARE'. The post came from the floor's shared tablet. Somebody pressed 'publish' without looking past the smiles.

TARGET
the tablet

Suspects and clues

Alabaster Snowball (AS)

1. Alabaster Snowball was right next to a rug.
2. Alabaster Snowball was in the same room as Ribb Bonbowford.

Ribb Bonbowford (RB)

3. Ribb Bonbowford was by the east outer wall.

Santa (SA)

4. Santa was the only person in the room.
5. Santa was in the Print Room.

Jack Frost (JF)

6. Jack Frost was by the west outer wall.

Pepper Minstix (PM)

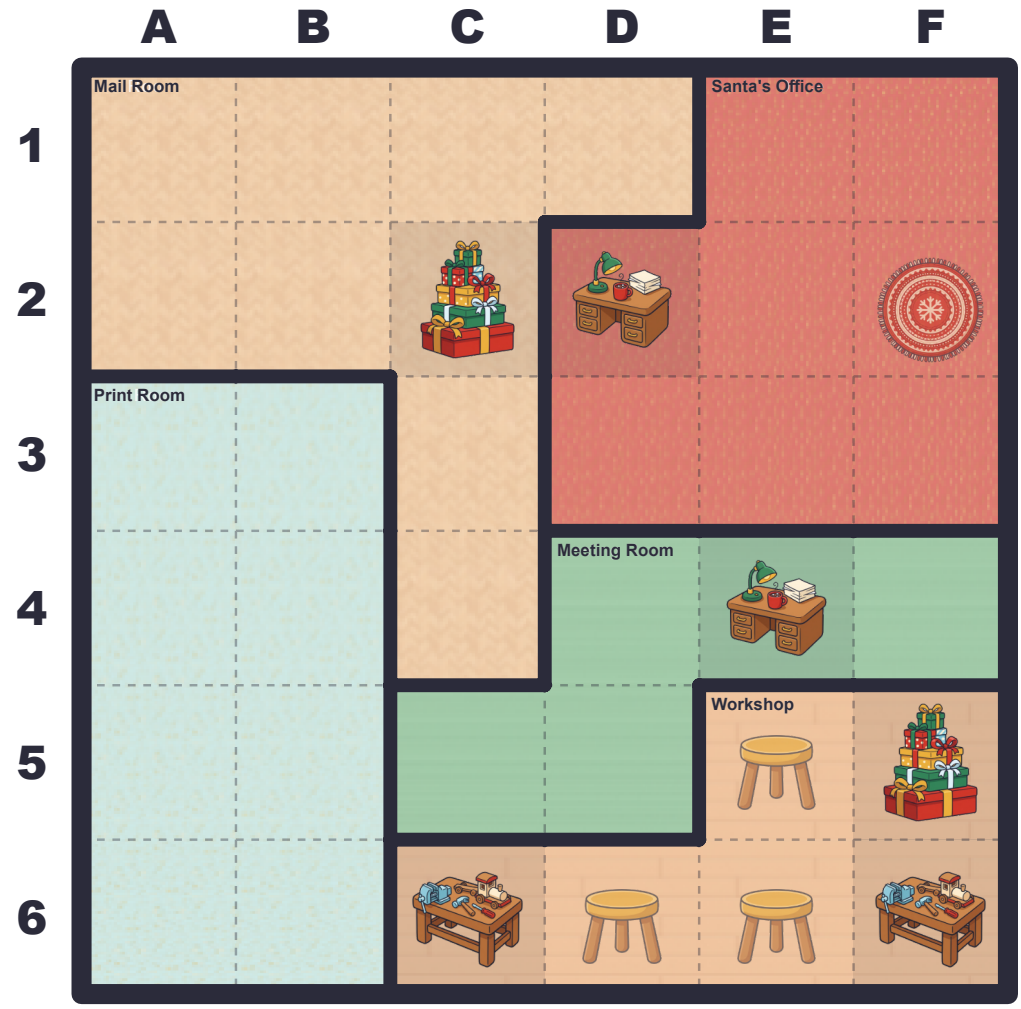
7. Pepper Minstix was further east than the tablet.
8. Pepper Minstix was in a room with a desk.
9. Pepper Minstix was further south than the tablet.

CASE 17

The Group Photo

Map of the floor · 6 × 6 · north is up

N



ROOMS

Meeting Room · Santa's Office · Workshop · Mail Room · Print Room

BLOCKED: NOBODY STANDS HERE

gift pile · desk · workbench

CAN BE STOOD ON

rug · stool

Notes

WHO	POSSIBLE SQUARES
AS Alabaster Snowball	
RB Ribb Bonbowford	
SA Santa	
JF Jack Frost	
PM Pepper Minstix	
T target	

Code grid

	A	B	C	D	E	F
1	r	A	3	r	V	9
2	Y	U	W	P	y	V
3	3	a	H	S	t	P
4	3	E	b	h	Q	X
5	r	P	C	n	N	U
6	R	Q	K	K	A	8

The character on the culprit's square is character **17** of the password.

Culprit _____ **No. 17**

CASE 18

LEVEL 3 · ANALYST

Free North Pole WiFi

A small WiFi dongle appeared at the back of a computer overnight. By morning it was offering a network called 'FreeNorthPoleWiFi' with an excellent signal and no password. Several devices joined at once, because free is free. Nobody asked who had installed it, why the workshop would suddenly need a second network, or why its login page wanted a badge number and a mother's maiden name. Somebody plugged that dongle in. The label on it says 'Speed booster'.

TARGET
the WiFi dongle

Suspects and clues

Santa (SA)

1. Santa was in the east half of the floor.
2. Santa was in the same room as Angel Candysalt.

Jack Frost (JF)

3. Jack Frost was further east than Santa.
4. Jack Frost was in the south half of the floor.
5. Jack Frost was further north than the WiFi dongle.
6. Jack Frost was further west than Alabaster Snowball.

Angel Candysalt (AC)

7. Angel Candysalt was further south than Santa.

Alabaster Snowball (AS)

No clue of their own. Look at the others.

Piney Sappington (PS)

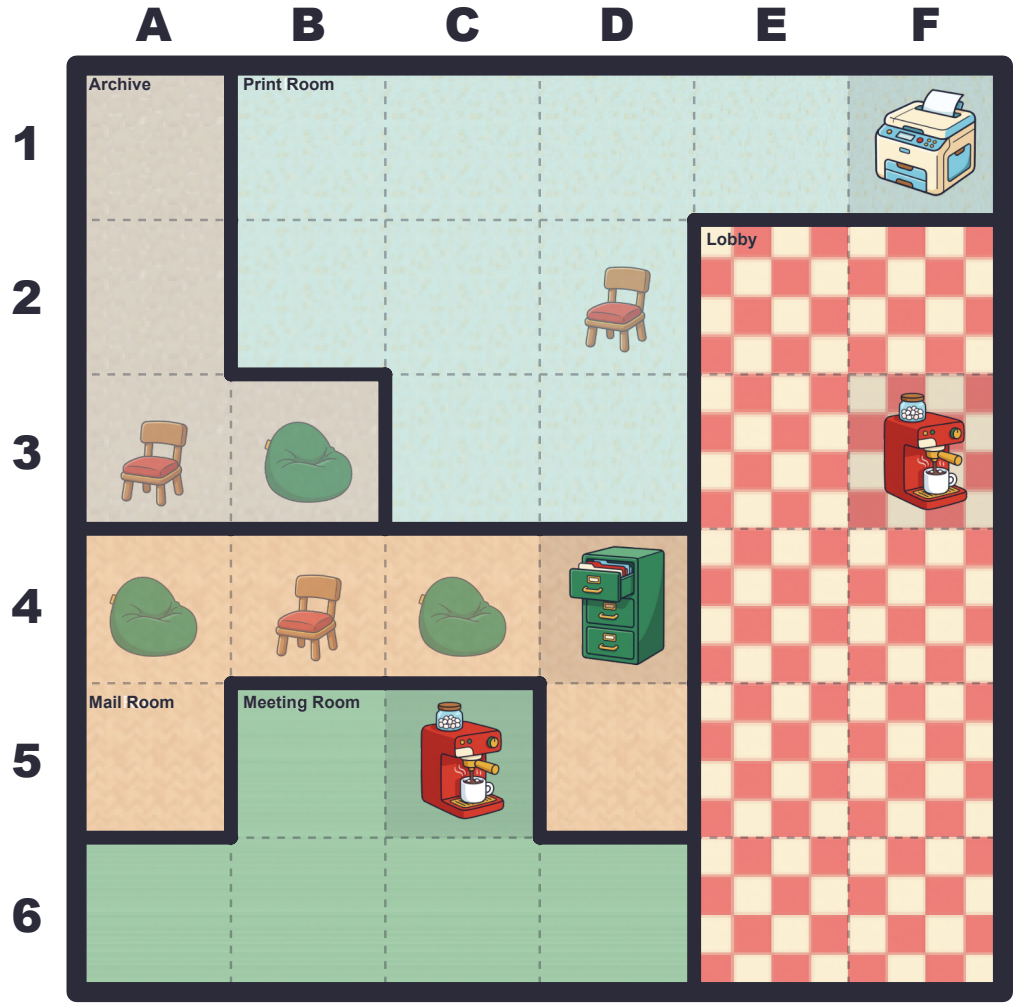
8. Piney Sappington was in a room with a cocoa machine.
9. Piney Sappington was further west than the WiFi dongle.

CASE 18

Free North Pole WiFi

Map of the floor · 6 × 6 · north is up

N



ROOMS

Archive · Lobby · Print Room · Mail Room · Meeting Room

BLOCKED: NOBODY STANDS HERE

printer · cocoa machine · filing cabinet

CAN BE STOOD ON

chair · beanbag

Notes

WHO	POSSIBLE SQUARES
SA Santa	
JF Jack Frost	
AC Angel Candysalt	
AS Alabaster Snowball	
PS Piney Sappington	
T target	

Code grid

	A	B	C	D	E	F
1	r	Z	f	T	3	i
2	B	2	P	C	B	2
3	g	X	b	m	i	r
4	R	K	7	W	A	L
5	B	A	6	d	5	n
6	B	Q	R	3	4	f

The character on the culprit's square is character **18** of the password.

Culprit _____ No. 18 ☐

THE STORY SO FAR

Chapter 4: The Budget

Eighteen characters, and something unheard of has happened: the security budget has been approved. Overnight the floor has acquired badge readers on the doors, colored access badges for everyone, door logs and a number of cameras, which the elves wave at.

It comes not a moment too soon. The incidents are no longer about forgotten printouts. Lists are being exported, cameras paused, firmware replaced. Somebody is working through the building with a plan.

Santa promotes you to Investigator and hands you a badge of your own. It is Gold. Jack Frost congratulates you warmly and asks, in passing, which rooms a Gold badge can open.

LEVEL 4

Investigator

CASES	MAP	SUSPECTS	NEW
19 - 24	7 × 7	6	Badges, cameras, door logs

How to solve this level

The map is now seven by seven with six people, and the new equipment gives you three new kinds of information. Badges: each person may only be in the rooms cleared for their badge color, so strike the forbidden rooms first. Cameras: each camera watches the squares listed for it, and a clue says whom it recorded, whom it did not, or how many. Door logs: the exact number of people in a room. Counting clues cut both ways: once a room is full, everybody else is out. And watch for pairs: if two people can only be in the same two rows, nobody else is in those rows. Guide: chapter 9.

The vault password so far

Write down the character from every solved case. All thirty have to be right.

LEVEL 1						LEVEL 2						LEVEL 3						LEVEL 4						LEVEL 5					
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30

CASE 19

LEVEL 4 · INVESTIGATOR

The Blind Spot

The camera console controls every camera on the floor. Last night somebody used it to make a handful of cameras gaze politely at the ceiling for eleven minutes - exactly the eleven minutes that mattered. The good news is that nobody thought to pause the remaining cameras, and that badge readers do not accept bribes. Between the footage that still exists and the access rights, the truth is somewhere on this page. Whoever did it was clearly hoping you would not look.

TARGET
the camera console

Suspects and clues

Angel Candysalt — AC

Red badge

1. Angel Candysalt was in the east half of the floor.
2. Angel Candysalt was in a room with a cocoa machine.
3. Angel Candysalt was further west than the camera console.

Eve Snowshoes — ES

Green badge

4. Camera C2 recorded Eve Snowshoes.
5. Eve Snowshoes was further east than Piney Sappington.

Piney Sappington — PS

Gold badge

6. Camera C1 did not record Piney Sappington.

Santa — SA

Green badge

7. Santa was further north than Angel Candysalt.

Bow Ninecandle — BN

Gold badge

No clue of their own. Look at the others.

Jack Frost — JF

Red badge

8. Jack Frost was further south than the camera console.

More clues

9. Camera C3 recorded nobody.
10. Camera C2 recorded exactly 2 people.
11. Door log Server Room: exactly 2 people in the room.

Access rights

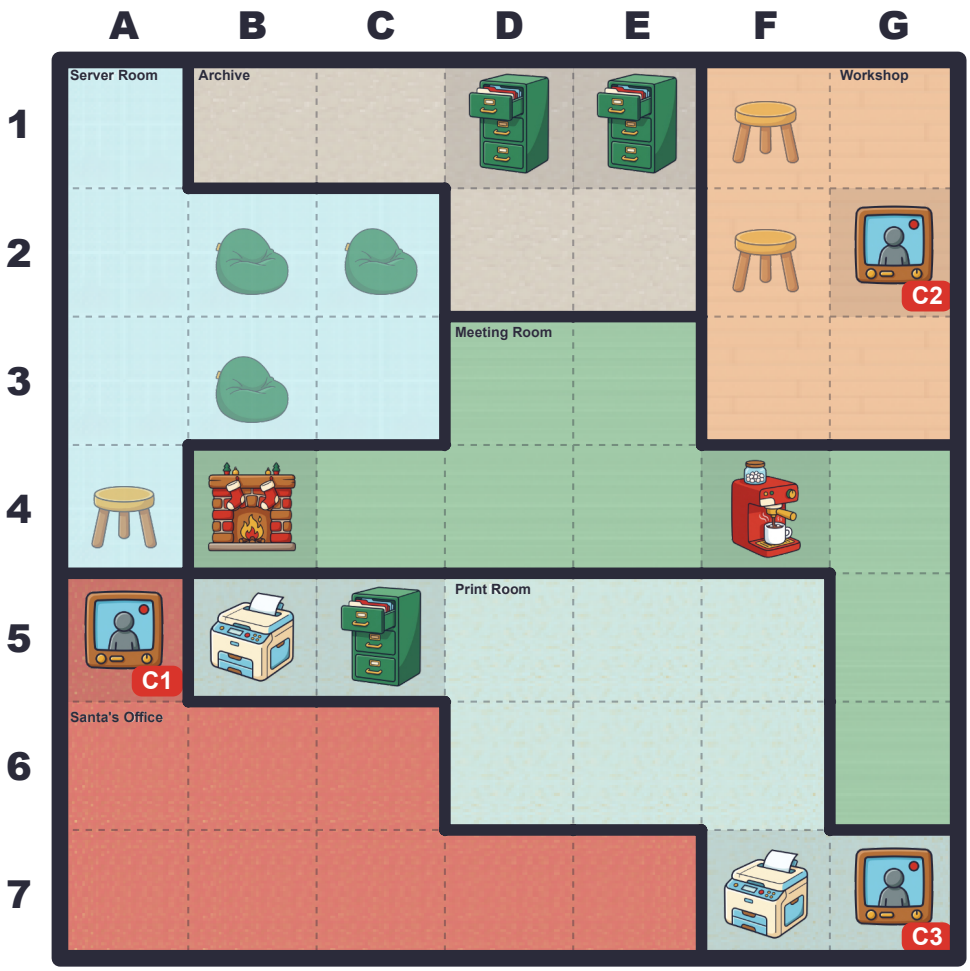
Red badge	Print Room, Meeting Room
Green badge	Server Room, Archive
Gold badge	Workshop, Santa's Office

CASE 19

The Blind Spot

Map of the floor · 7 × 7 · north is up

N



ROOMS

Server Room · Workshop · Archive · Print Room · Meeting Room · Santa's Office

BLOCKED: NOBODY STANDS HERE

filing cabinet · fireplace · cocoa machine · printer

CAN BE STOOD ON

stool · beanbag

CAMERAS (BLOCKED SQUARES)

C1 at A5 · watches A1 A2 A3 A4 A6 A7

C2 at G2 · watches G1 A2 B2 C2 D2 E2 F2 G3 G4 G5 G6

C3 at G7 · watches G3 G4 G5 G6

Notes

WHO	POSSIBLE SQUARES
AC Angel Candysalt	
ES Eve Snowshoes	
PS Piney Sappington	
SA Santa	
BN Bow Ninecandle	
JF Jack Frost	
T target	

Code grid

	A	B	C	D	E	F	G
1	y	P	r	e	9	M	5
2	h	X	n	4	F	j	M
3	E	i	S	4	A	3	X
4	h	J	4	g	R	D	W
5	2	i	a	9	h	K	h
6	U	W	9	X	j	H	4
7	d	m	3	3	M	M	L

The character on the culprit's square is character **19** of the password.

Culprit _____ No. 19

CASE 20

LEVEL 4 · INVESTIGATOR

The Gift List Export

The gift tracker knows where every single present is, who it is for and what is inside. It is, in other words, the one file a certain kind of person would dearly love to own. Last night somebody pressed 'Export all', waited patiently through the progress bar and walked out with the complete list. The dialog even asked 'Are you sure?'. They were sure. The door logs are quite clear about how many people were where. The people themselves are being rather less clear.

TARGET
the gift tracker

Suspects and clues

Alabaster Snowball — AS

Green badge

1. Camera C1 did not record Alabaster Snowball.
2. Alabaster Snowball was on a stool.

Ginger Breddie — GB

Gold badge

3. Camera C3 did not record Ginger Breddie.

Bow Ninecandle — BN

Red badge

4. Bow Ninecandle was in a room with a gift pile.

Angel Candysalt — AC

Green badge

5. Angel Candysalt was in the west half of the floor.

Jack Frost — JF

Red badge

6. Camera C1 recorded Jack Frost.

Santa — SA

Gold badge

7. Santa was right next to a beanbag.

More clues

8. The gift tracker was further north than Jack Frost.
9. Camera C3 recorded exactly 1 person.
10. Door log Mail Room: exactly 1 person in the room.

Access rights

Red badge	Storage Room, Santa's Office, Mail Room, Archive
Green badge	Storage Room, Archive, Lobby
Gold badge	Server Room, Santa's Office, Mail Room

CASE 20

The Gift List Export

Map of the floor · 7 × 7 · north is up

N

ROOMS

Storage Room · Server Room · Santa's Office · Mail Room · Archive · Lobby

BLOCKED: NOBODY STANDS HERE

desk · server rack · gift pile · filing cabinet

CAN BE STOOD ON

stool · beanbag

CAMERAS (BLOCKED SQUARES)

C1 at D2 · watches D1 E2 F2 G2 D3 D4 D5

C2 at G1 · watches A1 B1 C1 D1 E1 F1 G2 G3 G4 G5 G6 G7

C3 at C7 · watches C6 A7 B7 D7

Notes

WHO	POSSIBLE SQUARES
AS Alabaster Snowball	
GB Ginger Breddie	
BN Bow Ninecandle	
AC Angel Candysalt	
JF Jack Frost	
SA Santa	
T target	

Code grid

	A	B	C	D	E	F	G
1	A	d	C	E	2	C	X
2	S	q	L	V	q	Q	q
3	S	6	d	V	m	y	j
4	8	B	i	a	G	m	7
5	B	V	t	d	5	L	W
6	C	d	M	N	6	g	B
7	N	d	b	V	9	a	4

The character on the culprit's square is character **20** of the password.

Culprit _____ No. 20

CASE 21

LEVEL 4 · INVESTIGATOR

Git Never Forgets

Three weeks ago somebody saved a private key into the toy-firmware repository by mistake, noticed, deleted it one commit later and considered the matter closed. It was not closed. Version control remembers everything, forever, like a grandmother at a family dinner. Last night someone dug the key out of the history and used it at the build terminal, shortly before Poinsettia Mcmittens got around to replacing it. The badges and the cameras narrow things down nicely.

TARGET
the build terminal

Suspects and clues

Ginger Breddie — GB

Red badge

1. Camera C1 did not record Ginger Breddie.

Santa — SA

Red badge

2. Camera C2 did not record Santa.
3. Camera C3 recorded Santa.
4. Santa was in the east half of the floor.

Jack Frost — JF

Green badge

No clue of their own. Look at the others.

Eve Snowshoes — ES

Green badge

5. Camera C1 recorded Eve Snowshoes.
6. Eve Snowshoes was further east than Ginger Breddie.

Pepper Minstix — PM

Gold badge

No clue of their own. Look at the others.

Poinsettia Mcmittens — PO

Gold badge

7. Camera C1 recorded Poinsettia Mcmittens.
8. Poinsettia Mcmittens was further south than Jack Frost.

More clues

9. Camera C3 recorded exactly 1 person.
10. Camera C2 recorded exactly 1 person.
11. Door log Storage Room: exactly 1 person in the room.
12. Door log Print Room: exactly 1 person in the room.

Access rights

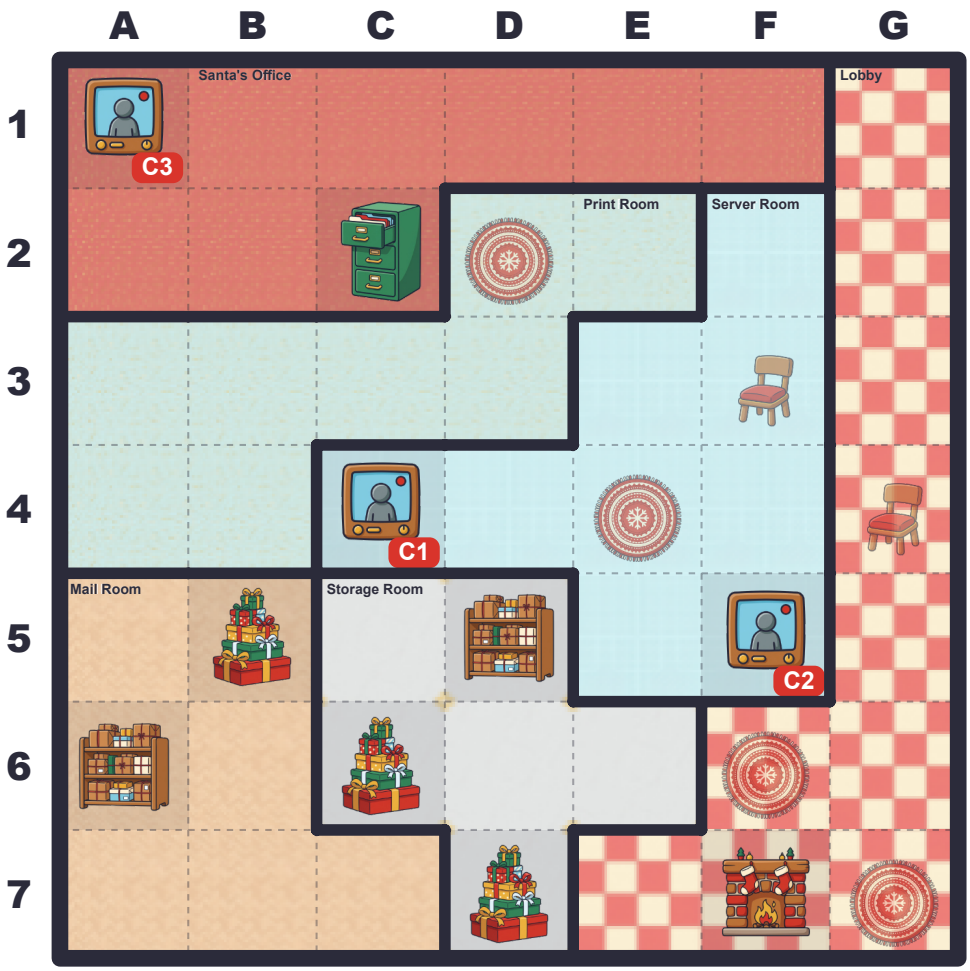
Red badge	Print Room, Storage Room, Santa's Office, Lobby
Green badge	Storage Room, Lobby
Gold badge	Mail Room, Server Room

CASE 21

Git Never Forgets

Map of the floor · 7 × 7 · north is up

N



ROOMS

Print Room · Mail Room · Storage Room · Santa's Office · Lobby · Server Room

BLOCKED: NOBODY STANDS HERE

filing cabinet · gift pile · parcel shelf · fireplace

CAN BE STOOD ON

rug · chair

CAMERAS (BLOCKED SQUARES)

C1 at C4 · watches C3 A4 B4 D4 E4 F4 G4 C5

C2 at F5 · watches F1 F2 F3 F4 E5 G5 F6

C3 at A1 · watches B1 C1 D1 E1 F1 G1 A2 A3 A4 A5

Notes

WHO	POSSIBLE SQUARES
GB Ginger Breddie	
SA Santa	
JF Jack Frost	
ES Eve Snowshoes	
PM Pepper Minstix	
PO Poinsettia Mcmittens	
T target	

Code grid

	A	B	C	D	E	F	G
1	b	C	t	Z	V	d	a
2	g	Y	e	K	3	R	A
3	D	q	6	e	5	g	F
4	N	f	K	a	G	V	2
5	r	L	Z	E	7	g	J
6	r	S	2	d	Z	S	R
7	t	i	W	3	i	Z	a

The character on the culprit's square is character **21** of the password.

Culprit _____ No. 21

CASE 22

LEVEL 4 · INVESTIGATOR

Shared with Everyone

The delivery schedule lives in a shared folder marked 'team only'. At some point yesterday the setting changed to 'anyone with the link', which in practice means anyone with an internet connection and mild curiosity. The change was made from the admin laptop. Everybody questioned so far has given the same answer: they never touch the admin laptop, they do not even know where it is kept, and in any case it was already like that when they arrived.

TARGET
the admin laptop

Suspects and clues

Piney Sappington

PS

Gold badge

1 Camera C1 recorded Piney Sappington.

Jack Frost

JF

Green badge

2 Jack Frost was on a beanbag.

Ginger Breddie

GB

Gold badge

3 Ginger Breddie was in the same room as Garland Candlesticks.

Garland Candlesticks

GC

Red badge

4 Camera C1 did not record Garland Candlesticks.

Morcel Nougat

MN

Green badge

5 Camera C3 recorded Morcel Nougat.

Santa

SA

Red badge

6 Camera C1 did not record Santa.

7 Santa was further south than the admin laptop.

More clues

8 Camera C3 recorded exactly 1 person.

Access rights

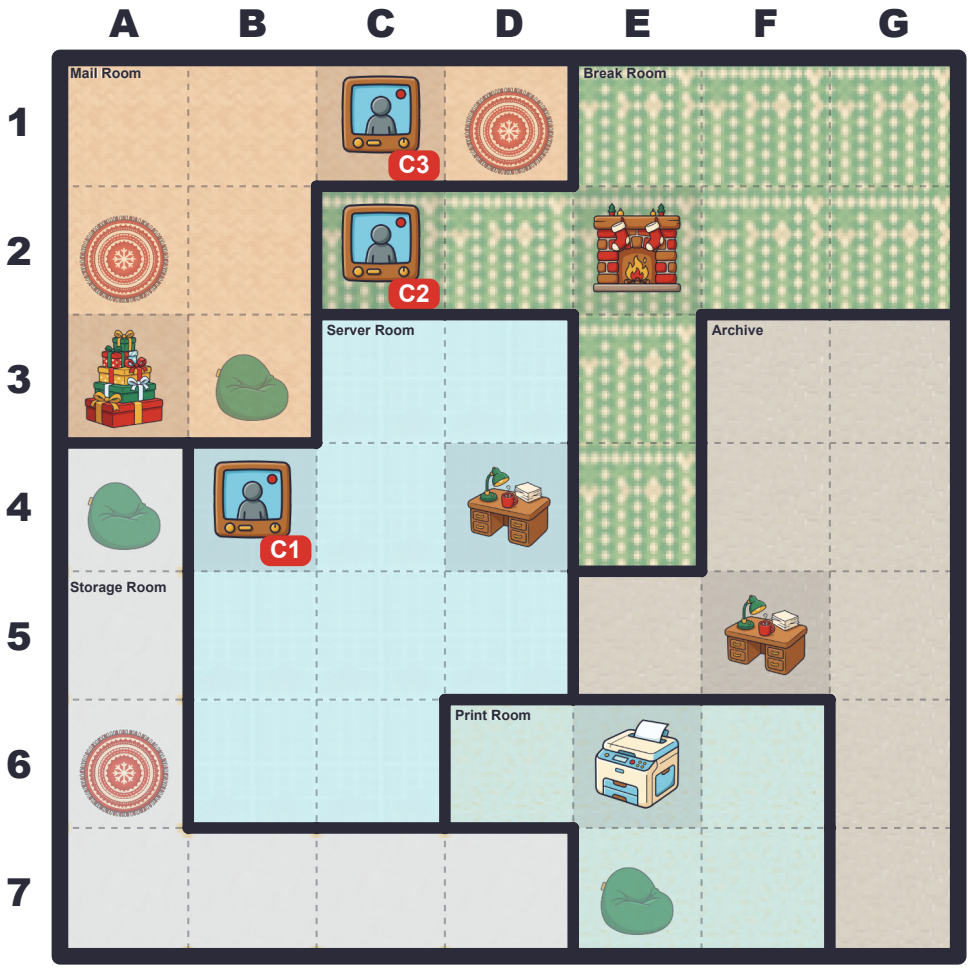
Red badge	Archive, Mail Room
Green badge	Archive, Print Room, Break Room, Storage Room
Gold badge	Archive, Server Room, Print Room, Storage Room, Mail Room

CASE 22

Shared with Everyone

Map of the floor · 7 × 7 · north is up

N



ROOMS

Archive · Server Room · Print Room · Break Room · Storage Room · Mail Room

BLOCKED: NOBODY STANDS HERE

fireplace · gift pile · desk · printer

CAN BE STOOD ON

rug · beanbag

CAMERAS (BLOCKED SQUARES)

C1 at B4 · watches B1 B2 B3 A4 C4 B5 B6 B7

C2 at C2 · watches A2 B2 D2 C3 C4 C5 C6 C7

C3 at C1 · watches A1 B1 D1 E1 F1 G1

Notes

WHO	POSSIBLE SQUARES
PS Piney Sappington	
JF Jack Frost	
GB Ginger Breddie	
GC Garland Candlesticks	
MN Morcel Nougat	
SA Santa	
T target	

Code grid

	A	B	C	D	E	F	G
1	j	R	E	L	q	g	6
2	C	5	h	X	D	X	6
3	M	9	2	A	b	V	R
4	h	i	6	7	e	6	Z
5	S	Q	8	K	A	Q	m
6	C	M	a	h	X	H	3
7	B	T	L	Y	N	f	n

The character on the culprit's square is character **22** of the password.

Culprit _____ No. 22

CASE 23

LEVEL 4 · INVESTIGATOR

The Label Maker Remembers

The label maker has one job: it prints address labels. Last week it received a firmware update that nobody ordered, from a website nobody recognizes, and since then it has taken on a second job. It quietly keeps a copy of every address it prints, which at this time of year means every address there is. Somebody stood at that machine and pressed 'Install' on a pop-up that said 'URGENT DRIVER FIX!!!'. Three exclamation marks. That alone should have been a clue.

TARGET
the label maker

Suspects and clues

Santa — SA

Red badge

1. Camera C1 recorded Santa.

Morcel Nougat — MN

Green badge

2. Morcel Nougat was in the south half of the floor.

Garland Candlesticks — GC

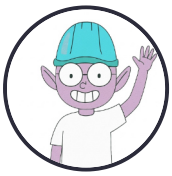
Red badge

3. Garland Candlesticks was on a rug.

Chimney Scissorsticks — CS

Gold badge

4. Chimney Scissorsticks was in a room with a rug.



Tinsel Upatree — TU

Green badge

5. Tinsel Upatree was the only person in the room.

Jack Frost — JF

Gold badge

6. Camera C3 recorded Jack Frost.
7. Jack Frost was further east than the label maker.

More clues

8. Camera C3 recorded exactly 1 person.

Access rights

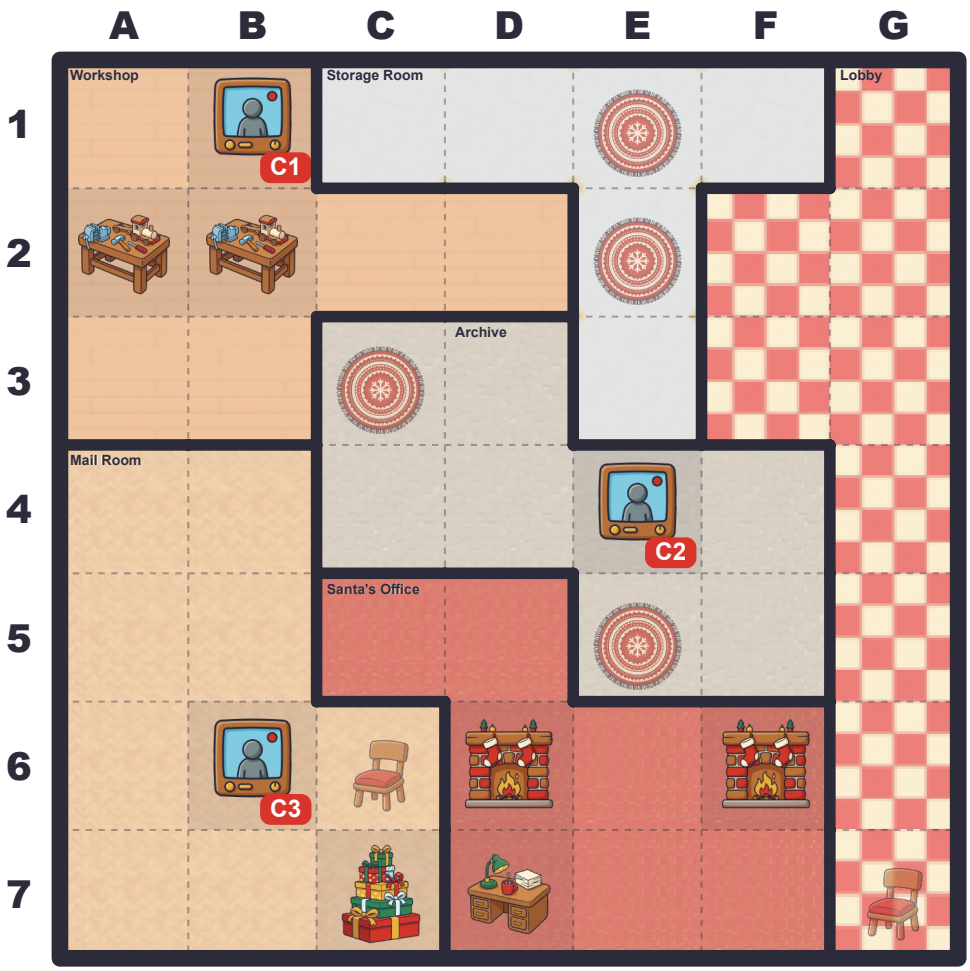
Red badge	Storage Room, Lobby
Green badge	Archive, Mail Room, Workshop
Gold badge	Archive, Mail Room, Workshop, Santa's Office

CASE 23

The Label Maker Remembers

Map of the floor · 7 × 7 · north is up

N



ROOMS

Archive · Storage Room · Mail Room · Lobby · Workshop · Santa's Office

BLOCKED: NOBODY STANDS HERE

workbench · fireplace · gift pile · desk

CAN BE STOOD ON

rug · chair

CAMERAS (BLOCKED SQUARES)

C1 at B1 · watches A1 C1 D1 E1 F1 G1

C2 at E4 · watches E1 E2 E3 A4 B4 C4 D4 F4 G4 E5 E6 E7

C3 at B6 · watches B3 B4 B5 A6 C6 B7

Notes

WHO	POSSIBLE SQUARES
SA Santa	
MN Morcel Nougat	
GC Garland Candlesticks	
CS Chimney Scissorsticks	
TU Tinsel Upatree	
JF Jack Frost	
T target	

Code grid

	A	B	C	D	E	F	G
1	Z	X	G	A	4	j	A
2	7	5	j	W	i	e	S
3	4	2	q	4	8	R	3
4	Y	K	J	d	K	3	b
5	C	L	i	M	G	G	J
6	d	Y	4	A	K	b	4
7	h	R	5	m	g	9	q

The character on the culprit's square is character **23** of the password.

Culprit _____ No. 23

CASE 24

LEVEL 4 · INVESTIGATOR

Candy Canes for Everyone

The candy cane dispenser used to hand out one candy cane per badge swipe. Since last night it empties its entire stock onto the floor whenever anybody walks past, which the elves regard as an improvement and the cleaning crew does not. Someone opened the service menu between two security rounds and changed the settings. The service code is printed on the inside of the lid, so that part was hardly a challenge. Follow the badges and the camera footage.

TARGET
the candy cane dispenser

Suspects and clues

Jewel Loggins — JL

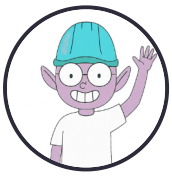
Gold badge

1. Camera C1 recorded Jewel Loggins.

Jack Frost — JF

Red badge

No clue of their own. Look at the others.



Tinsel Upatree — TU

Green badge

2. Tinsel Upatree was right next to a wrapping table.

Ribb Bonbowford — RB

Red badge

3. Camera C2 did not record Ribb Bonbowford.
4. Ribb Bonbowford was further north than Jack Frost.
5. Ribb Bonbowford was in a room with a server rack.

Alabaster Snowball — AS

Green badge

6. Alabaster Snowball was in the east half of the floor.

Santa — SA

Gold badge

7. Santa was right next to a rug.

More clues

8. Camera C3 recorded nobody.
9. Door log Print Room: exactly 2 people in the room.

Access rights

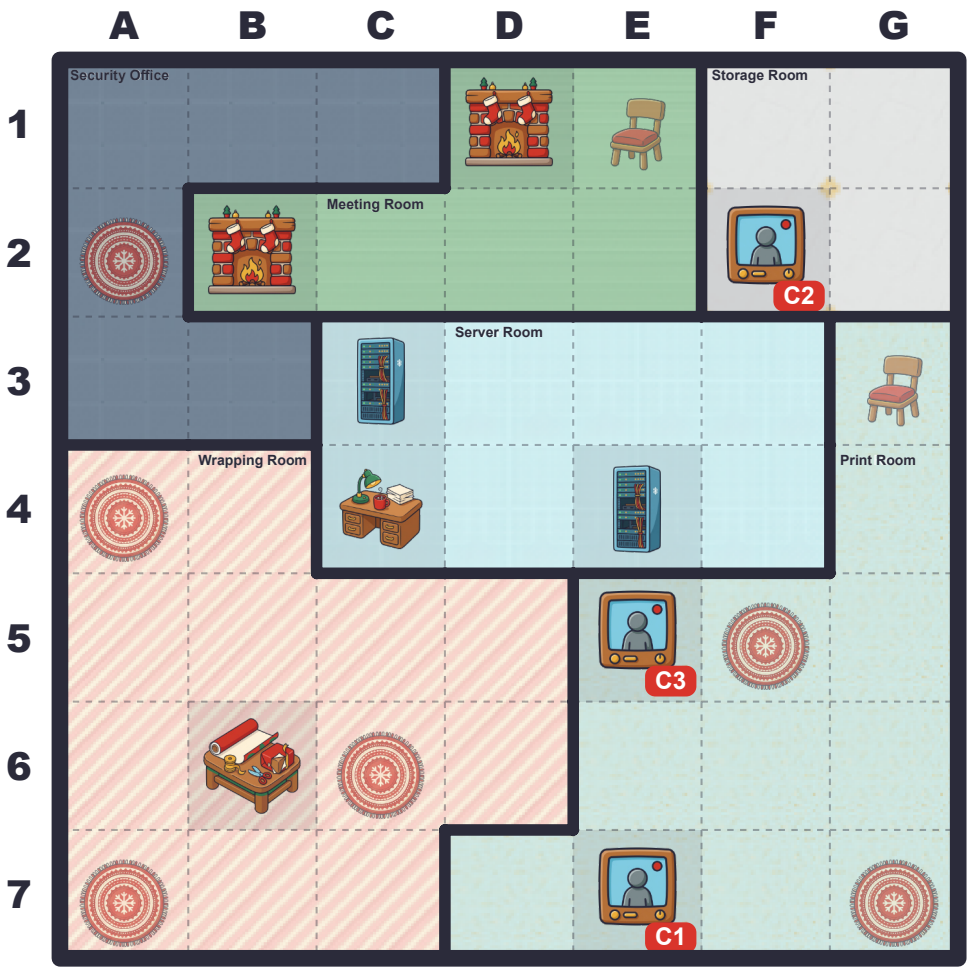
Red badge	Server Room, Wrapping Room, Meeting Room, Print Room
Green badge	Wrapping Room, Meeting Room
Gold badge	Wrapping Room, Security Office, Print Room, Storage Room

CASE 24

Candy Canes for Everyone

Map of the floor · 7 × 7 · north is up

N



ROOMS

Server Room · Wrapping Room · Security Office · Meeting Room · Print Room · Storage Room

BLOCKED: NOBODY STANDS HERE

fireplace · server rack · desk · wrapping table

CAN BE STOOD ON

chair · rug

CAMERAS (BLOCKED SQUARES)

C1 at E7 · watches E6 A7 B7 C7 D7 F7 G7

C2 at F2 · watches F1 C2 D2 E2 G2 F3 F4 F5 F6 F7

C3 at E5 · watches A5 B5 C5 D5 F5 G5 E6

Notes

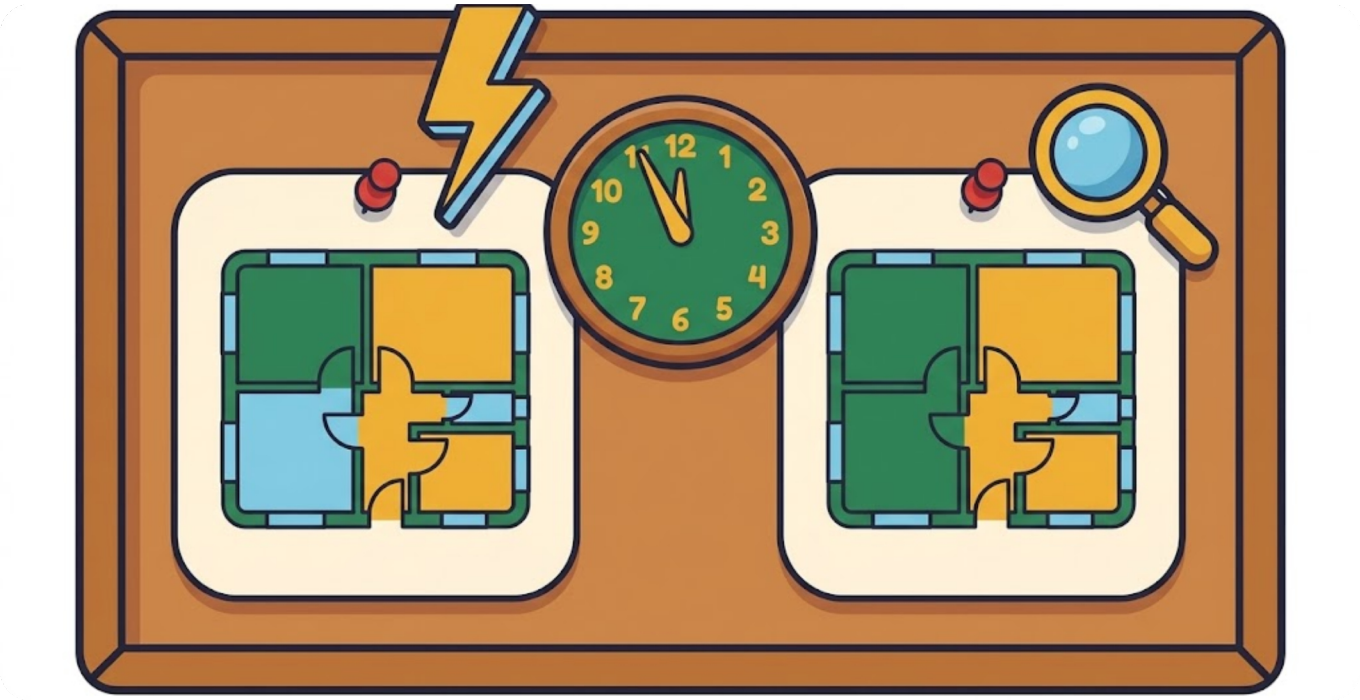
WHO	POSSIBLE SQUARES
JL Jewel Loggins	
JF Jack Frost	
TU Tinsel Upatree	
RB Ribb Bonbowford	
AS Alabaster Snowball	
SA Santa	
T target	

Code grid

	A	B	C	D	E	F	G
1	K	g	U	4	X	C	j
2	7	F	D	M	t	y	g
3	K	a	Y	U	4	a	Q
4	S	m	q	i	m	q	2
5	K	6	a	y	A	i	q
6	y	G	j	j	Z	9	7
7	T	n	K	F	8	C	d

The character on the culprit's square is character **24** of the password.

Culprit _____ No. 24



THE STORY SO FAR

Chapter 5: Everybody Lies a Little

Twenty-four characters. Christmas Eve is tomorrow, the vault is still shut, and the last six incidents are the worst yet: ransomware, forged keys, a blizzard indoors.

The mood on the floor has changed. Until now people left things out. Now, when they are guilty, they lie. One of them lies whether he is guilty or not: every word Jack Frost has said to you this week has turned out to be false, including 'good morning'. Santa, at least, remains incapable of it.

The new systems give you one advantage. For every incident you now have two pictures of the floor: the moment it happened and the moment it was discovered. Santa promotes you to Incident Commander. There is nobody left to be promoted above.

LEVEL 5

Incident Commander

CASES	MAP	SUSPECTS	NEW
25 - 30	6 × 6, two snapshots	5	Two snapshots and liars

How to solve this level

Each case has two grids, Incident and Discovery, and every person makes one statement. Santa always tells the truth, Jack Frost always lies, and everyone else lies only if they are the culprit. Start with what is certain: the evidence, Santa's statement, and the opposite of Jack's. Before you test anybody, check the statements against the evidence: an elf whose statement is already proven true cannot be the culprit. Then test the remaining suspects one by one. Assume a suspect is the culprit, turn that suspect's statement into its opposite, and fill in both grids until something breaks. Exactly one suspect survives. Guide: chapter 10.

The vault password so far

Write down the character from every solved case. All thirty have to be right.

LEVEL 1						LEVEL 2						LEVEL 3						LEVEL 4						**LEVEL 5**					
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30

CASE 25

LEVEL 5 · INCIDENT COMMANDER

Frozen Assets

This is the big one. The Naughty-Nice List is encrypted, and the list terminal shows a ransom note demanding 'an early spring, payable in small unmarked snowflakes'. Without the list nobody knows who gets coal, and guessing is bad for morale. You have two snapshots of the floor: who was where when the ransomware started, and who was where when it was discovered. Everyone has made a statement. As you will have come to expect, not all of them are true.

TARGET
the list terminal

Suspects, statements and evidence

Statements (speech bubbles): Santa always tells the truth. Jack Frost always lies. Everyone else tells the truth, except the culprit.
Evidence (numbered) is always true.

Jack Frost (JF)

DISCOVERY *"I was in a room with a fireplace."*

1. INCIDENT Jack Frost was right next to a filing cabinet.
2. DISCOVERY Jack Frost was right next to a gift pile.
3. DISCOVERY Jack Frost was in the same room as Santa.

Garland Candlesticks (GC)

INCIDENT *"I was in a room with a gift pile."*

4. DISCOVERY Garland Candlesticks was on a beanbag.

Santa (SA)

DISCOVERY *"I was further south than the list terminal."*

Bow Ninecandle (BN)

"I moved to a different room."

5. INCIDENT Bow Ninecandle was further west than Garland Candlesticks.

Piney Sappington (PS)

INCIDENT *"I was in a room with a fireplace."*

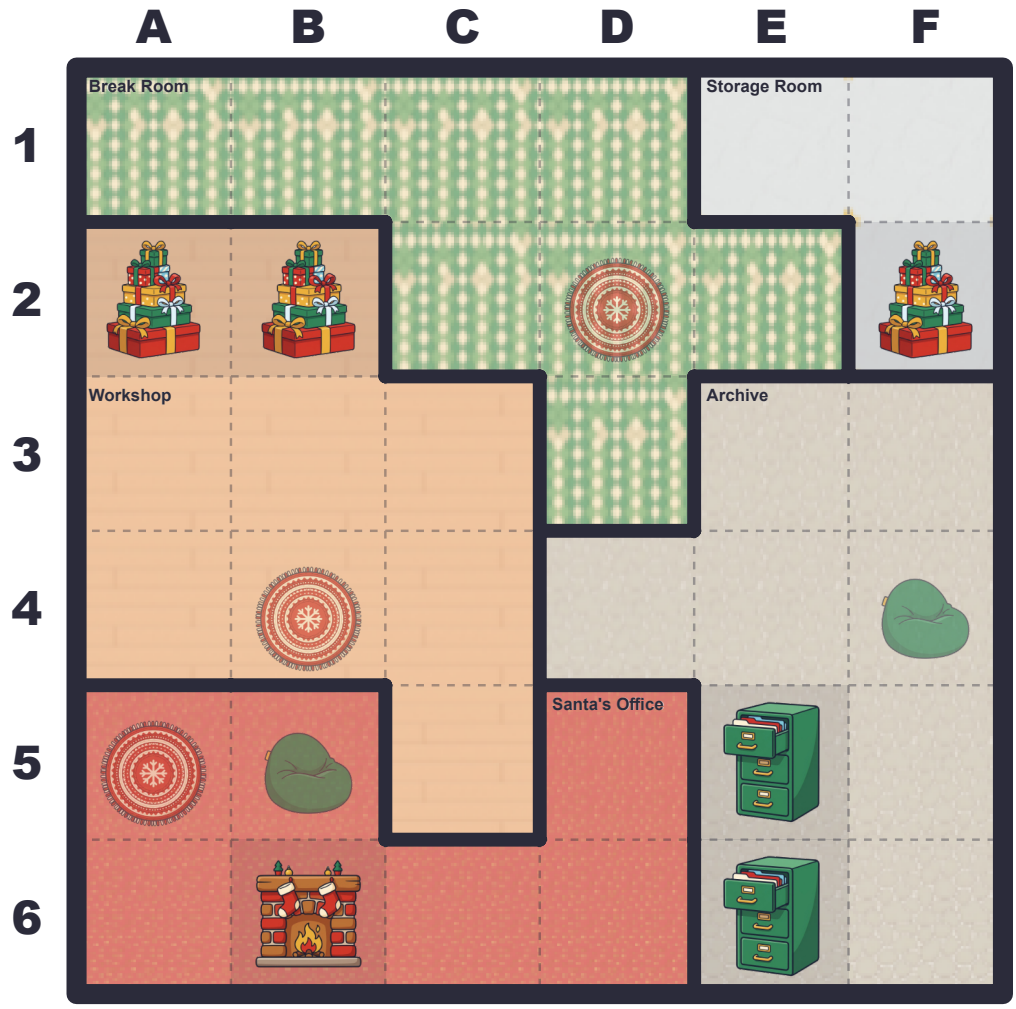
6. Piney Sappington moved to a different room.
7. INCIDENT Piney Sappington was on a beanbag.

CASE 25

Frozen Assets

Map of the floor · 6 × 6 · north is up

N



ROOMS

Santa's Office · Storage Room · Workshop · Break Room · Archive

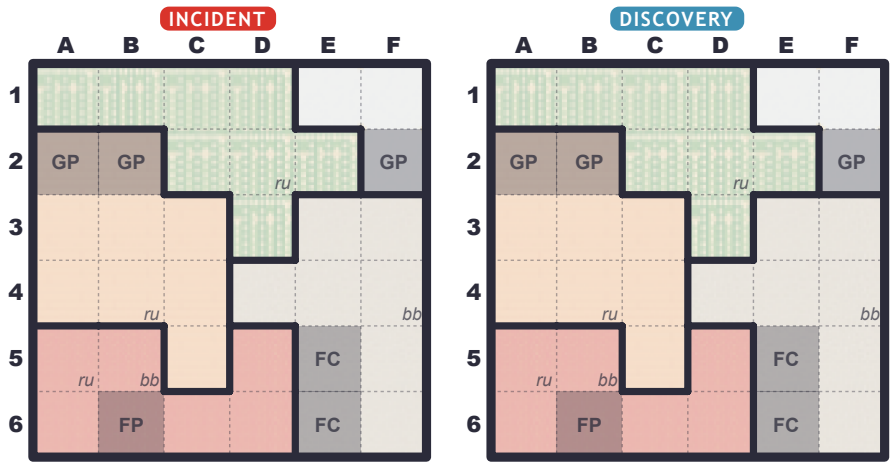
BLOCKED: NOBODY STANDS HERE

gift pile · filing cabinet · fireplace

CAN BE STOOD ON

rug · beanbag

Notes



Code grid

	A	B	C	D	E	F
1	n	B	n	2	D	F
2	f	W	n	f	Y	X
3	7	Y	C	B	d	g
4	h	L	R	S	h	n
5	S	M	8	N	h	j
6	j	e	m	E	W	7

The character on the culprit's square at the time of the incident is character **25** of the password.

Culprit _____ No. 25

CASE 26

LEVEL 5 · INCIDENT COMMANDER

Admin for a Night

The signing station issues the digital keys that open every system on the floor. Each request needs a form, a reason and a second signature. Last night it issued a key for the account 'admin' on the strength of a form whose reason field said 'because' and whose second signature was a drawing of a snowman. That key then opened everything. Two snapshots and a handful of statements are all you have. One of the people involved lies as a matter of principle.

TARGET
the signing station

Suspects, statements and evidence

Statements (speech bubbles): Santa always tells the truth. Jack Frost always lies. Everyone else tells the truth, except the culprit.
Evidence (numbered) is always true.

Morcel Nougat (MN)

INCIDENT *"I was right next to a wrapping table."*

1. DISCOVERY Morcel Nougat was further east than Wombley Cube.

Santa (SA)

INCIDENT *"I was not in the Break Room."*

2. INCIDENT Santa was right next to a chair.

Noel Boetie (NB)

INCIDENT *"I was not in the same room as Morcel Nougat."*

3. Noel Boetie moved to a different room.
4. INCIDENT Noel Boetie was further south than Jack Frost.
5. DISCOVERY Noel Boetie was in the north half of the floor.
6. DISCOVERY Noel Boetie was further south than the signing station.

Jack Frost (JF)

INCIDENT *"I was in the same room as Noel Boetie."*

7. INCIDENT Jack Frost was further west than Noel Boetie.

Wombley Cube (WC)

INCIDENT *"I was further south than Morcel Nougat."*

8. INCIDENT Wombley Cube was further west than Morcel Nougat.
9. DISCOVERY Wombley Cube was on a chair.

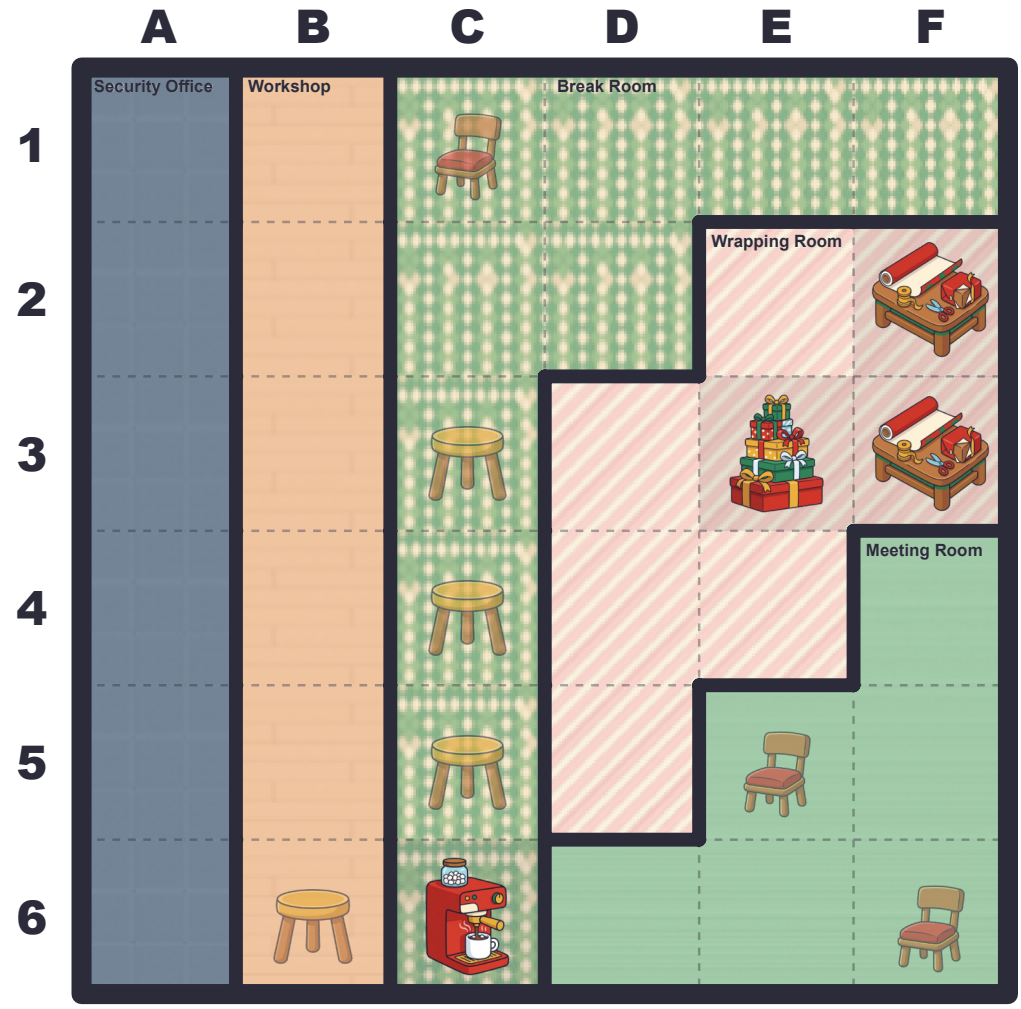
More evidence

10. INCIDENT Door log Meeting Room: exactly 2 people in the room.
11. DISCOVERY Door log Wrapping Room: exactly 1 person in the room.
12. DISCOVERY Door log Break Room: nobody in the room.

Admin for a Night

Map of the floor · 6 × 6 · north is up

N



ROOMS

Wrapping Room · Workshop · Meeting Room · Security Office · Break Room

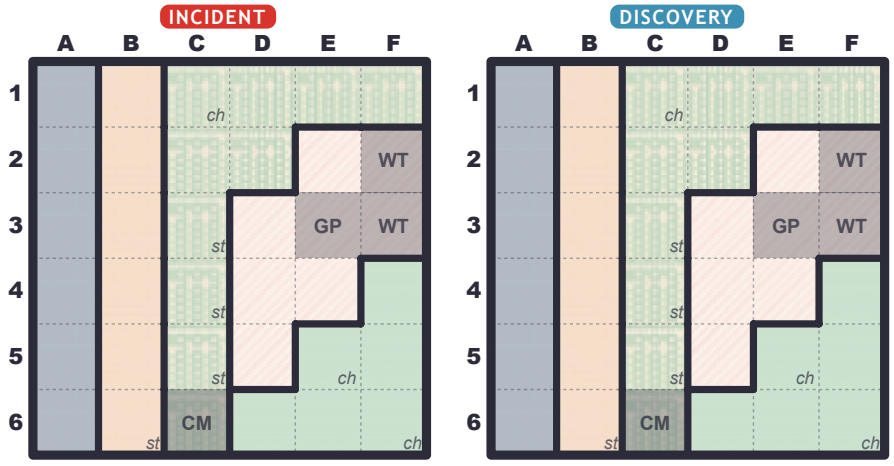
BLOCKED: NOBODY STANDS HERE

wrapping table · gift pile · cocoa machine

CAN BE STOOD ON

chair · stool

Notes



Code grid

	A	B	C	D	E	F
1	P	Y	Y	7	E	H
2	D	q	j	E	g	e
3	Z	H	5	E	S	W
4	2	S	V	b	t	E
5	9	j	5	E	n	N
6	8	H	2	C	K	2

The character on the culprit's square at the time of the incident is character **26** of the password.

Culprit _____ No. 26 ☐

CASE 27

LEVEL 5 · INCIDENT COMMANDER

The Big Blue Button

The snow machine exists to dust the windows with a tasteful amount of snow. It has two settings, 'festive' and 'blizzard', and one big blue button whose label wore off years ago. Last night somebody chose 'blizzard', pointed the nozzle indoors and pressed the button. By the time it was stopped, half the floor had a ski season. Everyone agrees it was dreadful. Nobody remembers being anywhere near it. Their statements are below, for whatever they are worth.

TARGET
the snow machine

Suspects, statements and evidence

Statements (speech bubbles): Santa always tells the truth. Jack Frost always lies. Everyone else tells the truth, except the culprit.
Evidence (numbered) is always true.

Santa (SA)

DISCOVERY *"I was in the Server Room."*

1. INCIDENT Santa was in a room with a fireplace.
2. DISCOVERY Santa was in the east half of the floor.
3. DISCOVERY Santa was further south than Poinsettia Mcmittens.

Alabaster Snowball (AS)

DISCOVERY *"I was right next to a Christmas tree."*

4. DISCOVERY Alabaster Snowball was in a room with a chair.
5. DISCOVERY Alabaster Snowball was further south than Wombley Cube.

Poinsettia Mcmittens (PO)

"I moved to a different room."

6. INCIDENT Poinsettia Mcmittens was further west than Wombley Cube.
7. DISCOVERY Poinsettia Mcmittens was right next to a stool.

Wombley Cube (WC)

INCIDENT *"I was not in the Meeting Room."*

8. Wombley Cube moved to a different room.

Jack Frost (JF)

INCIDENT *"I was right next to a fireplace."*

9. INCIDENT Jack Frost was right next to a chair.
10. INCIDENT Jack Frost was in the west half of the floor.
11. DISCOVERY Jack Frost was further north than the snow machine.

More evidence

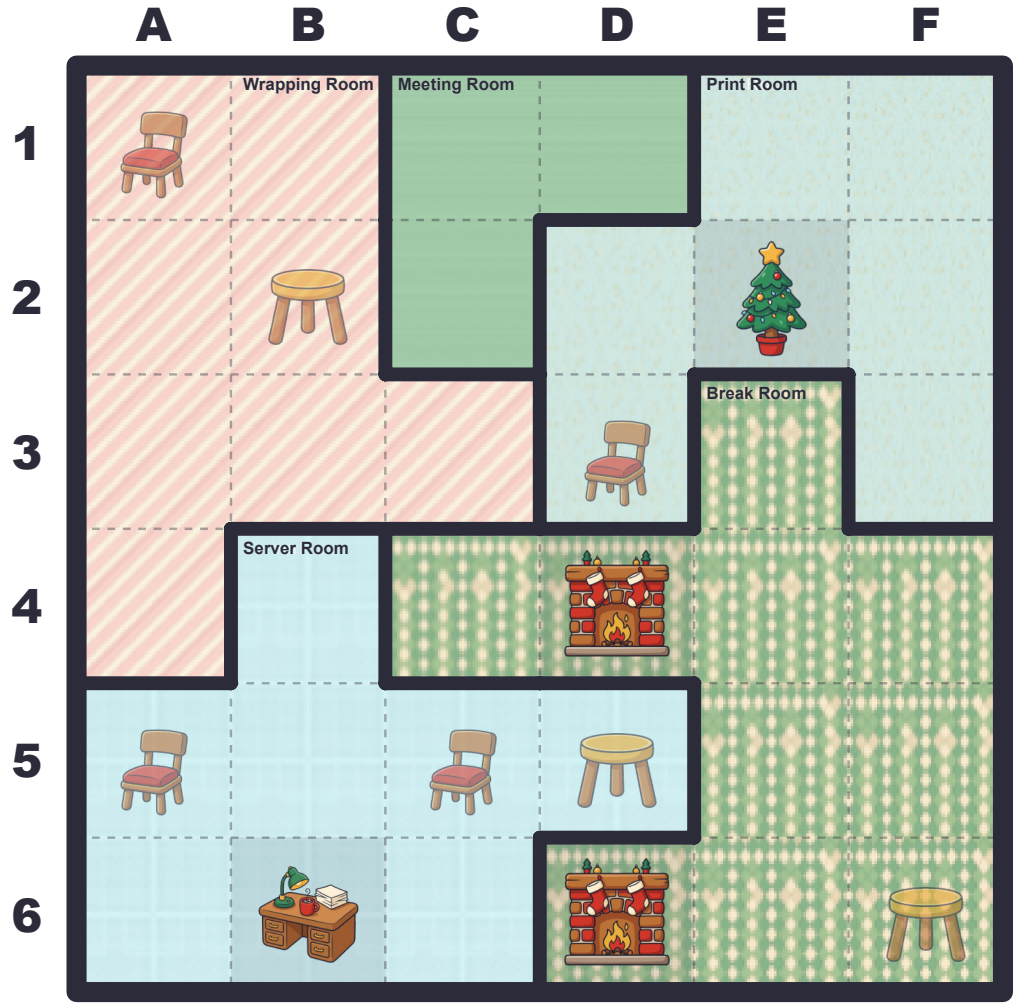
12. INCIDENT Door log Meeting Room: exactly 1 person in the room.
13. INCIDENT Door log Wrapping Room: exactly 1 person in the room.
14. DISCOVERY Door log Wrapping Room: exactly 1 person in the room.
15. DISCOVERY Door log Server Room: exactly 1 person in the room.

CASE 27

The Big Blue Button

Map of the floor · 6 × 6 · north is up

N



ROOMS

Break Room · Wrapping Room · Server Room · Meeting Room · Print Room

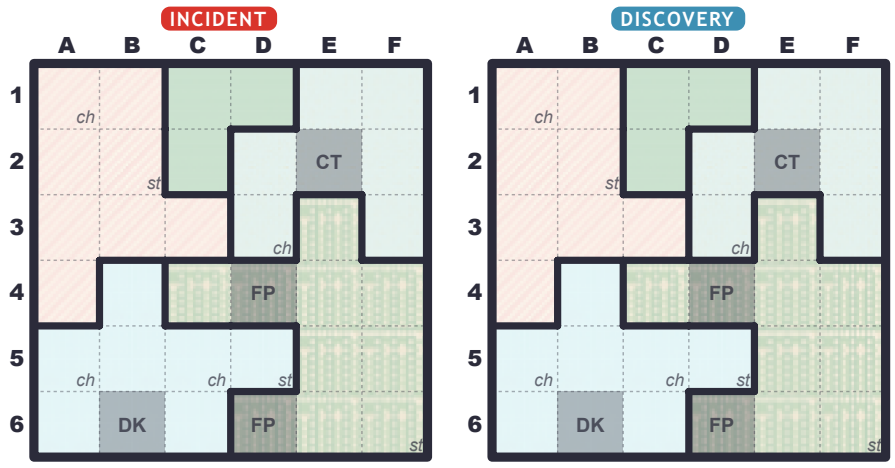
BLOCKED: NOBODY STANDS HERE

Christmas tree · fireplace · desk

CAN BE STOOD ON

chair · stool

Notes



Code grid

	A	B	C	D	E	F
1	P	4	B	J	T	P
2	R	Y	n	M	9	R
3	9	W	f	d	D	S
4	T	8	h	t	q	Y
5	H	T	Q	N	T	5
6	Q	r	P	m	q	G

The character on the culprit's square at the time of the incident is character **27** of the password.

Culprit _____ **No. 27** ☐

CASE 28

LEVEL 5 · INCIDENT COMMANDER

The Short Password

Somewhere on this floor there was a service account with the password 'elf' and permission to do almost anything. Fixing it had been on the to-do list since 2019. Last night somebody finally dealt with it, though not in the way the to-do list had in mind: the password was cracked in under a second, and three steps later the directory console and every account on the floor belonged to someone else. The trail is still warm. So, suspiciously, is the marshmallow on the keyboard.

TARGET
the directory console

Suspects, statements and evidence

Statements (speech bubbles): Santa always tells the truth. Jack Frost always lies. Everyone else tells the truth, except the culprit.
Evidence (numbered) is always true.

Jack Frost (JF)

"I stayed in the same room."

Santa (SA)

DISCOVERY *"I was in the north half of the floor."*

1. DISCOVERY Santa was right next to a beanbag.
2. DISCOVERY Santa was further east than Tinsel Upatree.
3. DISCOVERY Santa was further north than Noel Boetie.

Poinsettia Mcmittens (PO)

"I moved to a different room."

4. INCIDENT Poinsettia Mcmittens was right next to a stool.
5. DISCOVERY Poinsettia Mcmittens was on a beanbag.
6. DISCOVERY Poinsettia Mcmittens was in a room with a filing cabinet.

Noel Boetie (NB)

DISCOVERY *"I was in a room with a parcel shelf."*

7. INCIDENT Noel Boetie was further east than Jack Frost.
8. INCIDENT Noel Boetie was in the south half of the floor.
9. DISCOVERY Noel Boetie was further north than Jack Frost.

Tinsel Upatree (TU)

"I stayed in the same room."

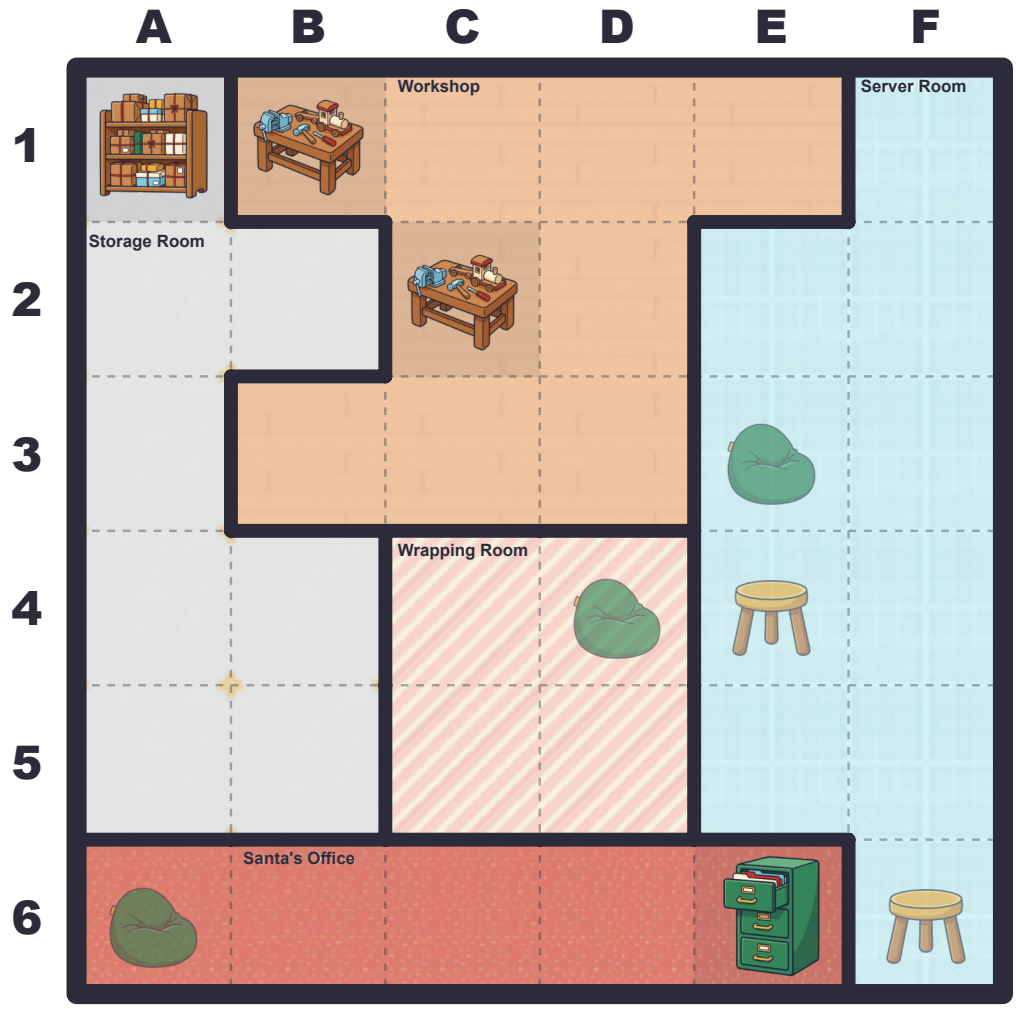
More evidence

10. DISCOVERY The directory console was further east than Jack Frost.
11. INCIDENT Door log Storage Room: exactly 2 people in the room.

The Short Password

Map of the floor · 6 × 6 · north is up

N



ROOMS

Server Room · Santa's Office · Storage Room · Wrapping Room · Workshop

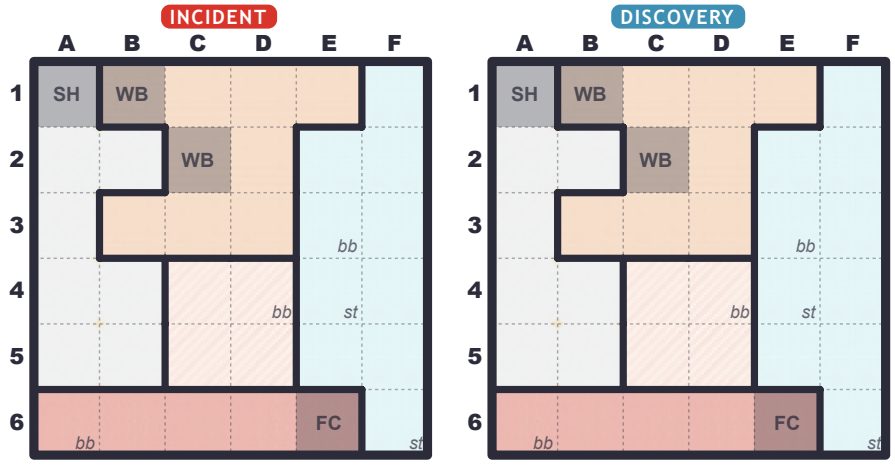
BLOCKED: NOBODY STANDS HERE

parcel shelf · workbench · filing cabinet

CAN BE STOOD ON

beanbag · stool

Notes



Code grid

	A	B	C	D	E	F
1	d	R	m	C	9	m
2	A	N	Z	G	C	B
3	Y	3	C	C	Q	U
4	h	3	q	Q	m	C
5	T	N	P	q	i	i
6	C	E	C	C	X	i

The character on the culprit's square at the time of the incident is character **28** of the password.

Culprit _____ **No. 28** ☐

CASE 29

LEVEL 5 · INCIDENT COMMANDER

The Guest List

Gift vouchers were strictly for invited guests. The voucher machine checks every name against the guest list, and it does so very thoroughly. What it does not check is who wrote the guest list. Someone handed it a list of their own making, on which they appear forty times under slightly different spellings, and the machine paid out forty times without a murmur. Two snapshots, one statement each. Choose carefully whom you believe.

TARGET
the voucher machine

Suspects, statements and evidence

Statements (speech bubbles): Santa always tells the truth. Jack Frost always lies. Everyone else tells the truth, except the culprit.
Evidence (numbered) is always true.

Santa — SA

"I moved to a different room."

1 DISCOVERY Santa was further south than the voucher machine.

Bow Ninecandle — BN

"I stayed in the same room."

2 INCIDENT Bow Ninecandle was right next to a filing cabinet.

Jingle Ringford — JR

DISCOVERY *"I was in the same room as Jack Frost."*

3 Jingle Ringford stayed in the same room.

Chimney Scissorsticks — CS

DISCOVERY *"I was not in the same room as Bow Ninecandle."*

4 DISCOVERY Chimney Scissorsticks was right next to a gift pile.

5 DISCOVERY Chimney Scissorsticks was further south than Santa.

Jack Frost — JF

INCIDENT *"I was in the west half of the floor."*

6 INCIDENT Jack Frost was further south than Chimney Scissorsticks.

More evidence

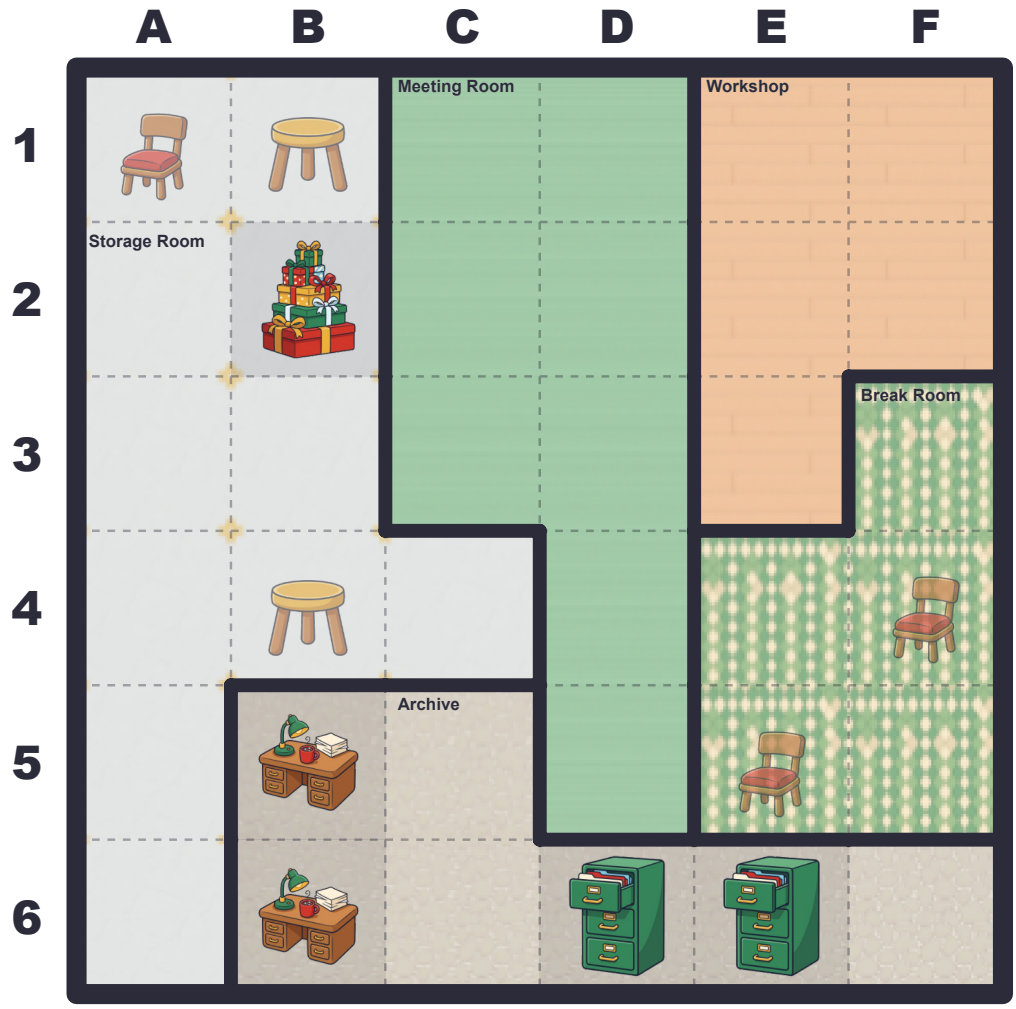
7 INCIDENT Door log Meeting Room: exactly 1 person in the room.

8 INCIDENT Door log Break Room: nobody in the room.

The Guest List

Map of the floor · 6 × 6 · north is up

N



ROOMS

Break Room · Meeting Room · Storage Room · Workshop · Archive

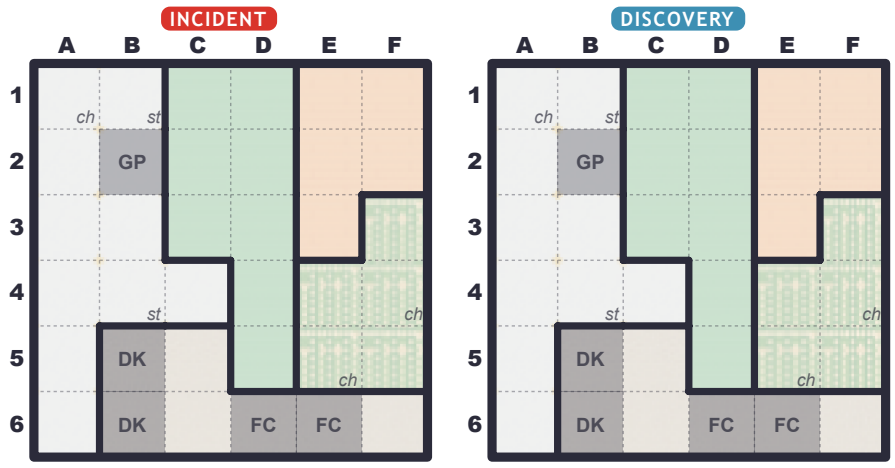
BLOCKED: NOBODY STANDS HERE

gift pile · desk · filing cabinet

CAN BE STOOD ON

chair · stool

Notes



Code grid

	A	B	C	D	E	F
1	A	U	H	X	C	T
2	7	r	t	t	2	8
3	q	4	g	7	b	F
4	e	7	g	b	J	L
5	b	4	C	K	Z	U
6	C	r	B	F	6	2

The character on the culprit's square at the time of the incident is character **29** of the password.

Culprit _____ **No. 29** ☐

CASE 30

LEVEL 5 · INCIDENT COMMANDER

Lights Out

At 4:17 in the afternoon the whole floor went offline: no mail, no gift tracker, no reindeer streaming. Someone had logged into the main router with the password printed on the sticker underneath it and installed an 'update' from a website with a snowflake logo. This is the last case. By now you may have noticed that one name appears on every single suspect list, and that honest mistakes on this floor tend to begin with a helpful suggestion. Look closely.

TARGET
the main router

Suspects, statements and evidence

Statements (speech bubbles): Santa always tells the truth. Jack Frost always lies. Everyone else tells the truth, except the culprit.
Evidence (numbered) is always true.

Garland Candlesticks (GC)

INCIDENT *"I was in the north half of the floor."*

1. Garland Candlesticks moved to a different room.
2. INCIDENT Garland Candlesticks was on a beanbag.
3. DISCOVERY Garland Candlesticks was in the west half of the floor.

Piney Sappington (PS)

DISCOVERY *"I was further north than Jack Frost."*

Jack Frost (JF)

DISCOVERY *"I was in the north half of the floor."*

4. INCIDENT Jack Frost was in the west half of the floor.

Santa (SA)

DISCOVERY *"I was in the west half of the floor."*

5. INCIDENT Santa was in the west half of the floor.
6. INCIDENT Santa was further south than Piney Sappington.
7. DISCOVERY Santa was right next to a Christmas tree.

Tinsel Upatree (TU)

INCIDENT *"I was in the Wrapping Room."*

8. DISCOVERY Tinsel Upatree was right next to a Christmas tree.

More evidence

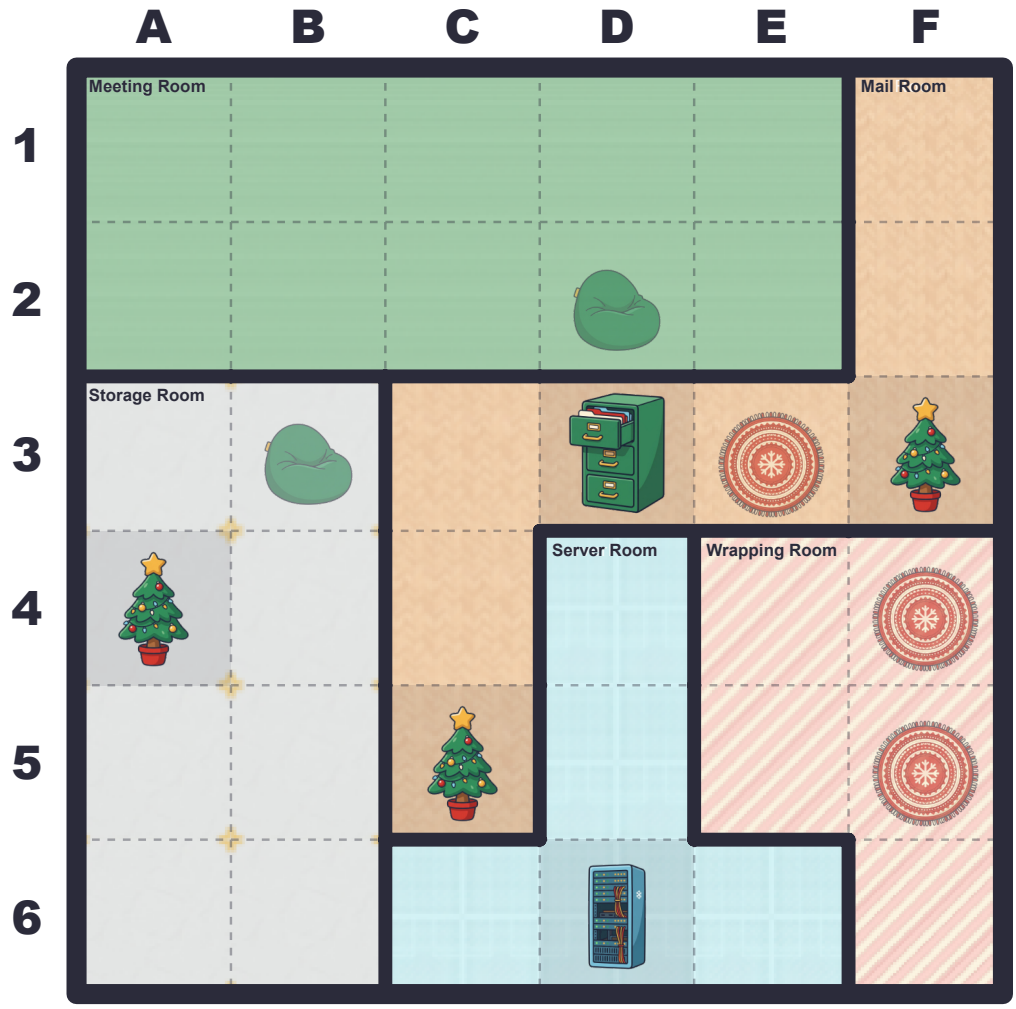
9. DISCOVERY The main router was further north than Santa.
10. INCIDENT Door log Mail Room: nobody in the room.
11. DISCOVERY Door log Mail Room: exactly 1 person in the room.
12. DISCOVERY Door log Server Room: exactly 1 person in the room.
13. DISCOVERY Door log Wrapping Room: nobody in the room.

CASE 30

Lights Out

Map of the floor · 6 × 6 · north is up

N



ROOMS

Mail Room · Server Room · Meeting Room · Wrapping Room · Storage Room

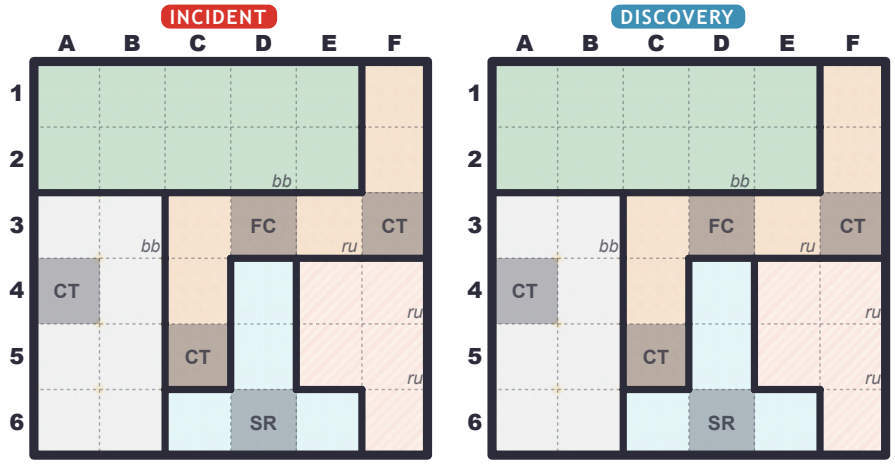
BLOCKED: NOBODY STANDS HERE

filing cabinet · Christmas tree · server rack

CAN BE STOOD ON

beanbag · rug

Notes



Code grid

	A	B	C	D	E	F
1	y	g	Z	4	R	a
2	N	G	N	e	P	P
3	A	2	M	7	Q	Q
4	5	L	6	W	D	Y
5	J	S	t	d	q	5
6	g	E	A	H	A	G

The character on the culprit's square at the time of the incident is character **30** of the password.

Culprit _____ **No. 30** ☐

THE END

Epilogue: Thirty Characters

Thirty cases, thirty characters. You type them into the keypad of the sleigh vault, slowly, because there is no second attempt. The door opens with the sigh of a machine that has been waiting to do this for three weeks.

Inside are the launch codes, untouched, and a deck chair. In the deck chair sits Jack Frost with a mug of something cold. He does not seem surprised. 'Most of the time,' he says, 'I never touched a thing. They did it themselves. I only made suggestions.'

He is not wrong, and that is the part that will keep you awake. Hardly any of the thirty incidents needed a master criminal. Most needed one helpful person and one moment of not thinking twice.

The sleigh leaves on time. The elves are not punished; they are trained, which several of them consider worse. The suggestion box is retired with full honors. Jack Frost receives a permanent place on the Naughty List and is enrolled in the security awareness course, front row, every session, for life. He has appealed against the second part.

The vault password

LEVEL 1 (1 2 3 4 5 6) LEVEL 2 (7 8 9 10 11 12) LEVEL 3 (13 14 15 16 17 18) LEVEL 4 (19 20 21 22 23 24) LEVEL 5 (25 26 27 28 29 30)